

Persuading voters using human–artificial intelligence dialogues

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There is great public concern about the potential use of generative artificial intelligence (AI) for political persuasion and the resulting impacts on elections and democracy^{1–6}. We inform these concerns using pre-registered experiments to assess the ability of large language models to influence voter attitudes. In the context of the 2024 US presidential election, the 2025 Canadian federal election and the 2025 Polish presidential election, we assigned participants randomly to have a conversation with an AI model that advocated for one of the top two candidates. We observed significant treatment effects on candidate preference that are larger than typically observed from traditional video advertisements^{7–9}. We also document large persuasion effects on Massachusetts residents' support for a ballot measure legalizing psychedelics. Examining the persuasion strategies⁹ used by the models indicates that they persuade with relevant facts and evidence, rather than using sophisticated psychological persuasion techniques. Not all facts and evidence presented, however, were accurate; across all three countries, the AI models advocating for candidates on the political right made more inaccurate claims. Together, these findings highlight the potential for AI to influence voters and the important role it might play in future elections.

There is deep public concern about the impact of artificial intelligence (AI) on democratic societies^{1–6}. Much of this concern focuses on recent advances in generative AI—such as the widespread availability of large language models (LLMs)—and the potential of this new technology to influence voter attitudes^{10–12}.

It remains unclear, however, how effective generative AI will be at actually changing voter attitudes, particularly in the context of highly information-saturated elections such as presidential races. Outside the context of AI, there is little evidence of substantial persuasion effects during such campaigns. For example, recent analyses of large numbers of field experiments⁷ and survey experiments^{8,9} in the USA found small average effects of political persuasion on candidate preference and choice on the order of one percentage point or smaller.

There is reason to expect AI-generated political advertisements will show similarly small effects. Recent studies indicate that when creating static persuasive messages, generative AI can sometimes match—but not typically exceed—the persuasive effect of human-generated content¹³, and due to scaling laws this result is unlikely to change even with much more sophisticated AI models¹⁴. From this perspective, it seems that concerns about the impact of generative AI on democracy—and in particular on presidential elections—may be overstated¹⁵.

A key aspect of the power of generative AI, however, is its ability to generate content in real time—and thus to engage in persuasive dialogues. Past work on persuasion in the context of prejudice reduction finds that conversations between canvassers and voters can have large and lasting effects^{16,17}. In the context of human–AI dialogues, there is evidence

that AI can outperform humans in persuasion¹⁸ and that dialogues can be more persuasive than static messages¹⁹. Furthermore, human–AI dialogues can also durably reduce conspiracy beliefs^{20–22}, which were previously thought to be largely impervious to evidence and arguments (much like political attitudes towards presidential candidates).

Thus, human–AI dialogues may have a substantive effect on voters' electoral decisions. Here, we investigate this question, asking whether dialogues with frontier AI models can change voters' political attitudes substantively and investigating the mechanisms underlying any such persuasion.

Persuasion in the US 2024 election

We began by investigating the effects of human–AI dialogues on attitudes towards presidential candidates and voting intentions in the 2024 US presidential election. We recruited 2,306 US citizens from Prolific and CloudConnect to engage in a conversation with an AI chatbot in late August and early September 2024. Participants indicated their preference for the Democratic candidate Kamala Harris versus the Republican candidate Donald Trump using a 0–100 scale, and their likelihood of voting in the election using a 0–100 scale. In addition to these pre-registered key outcomes, we also asked a categorical choice question, which we analysed post hoc, of how the participant would vote if the 2024 election happened today.

After completing these pre-treatment measures, participants were randomized first to either indicate the policy issue that was most

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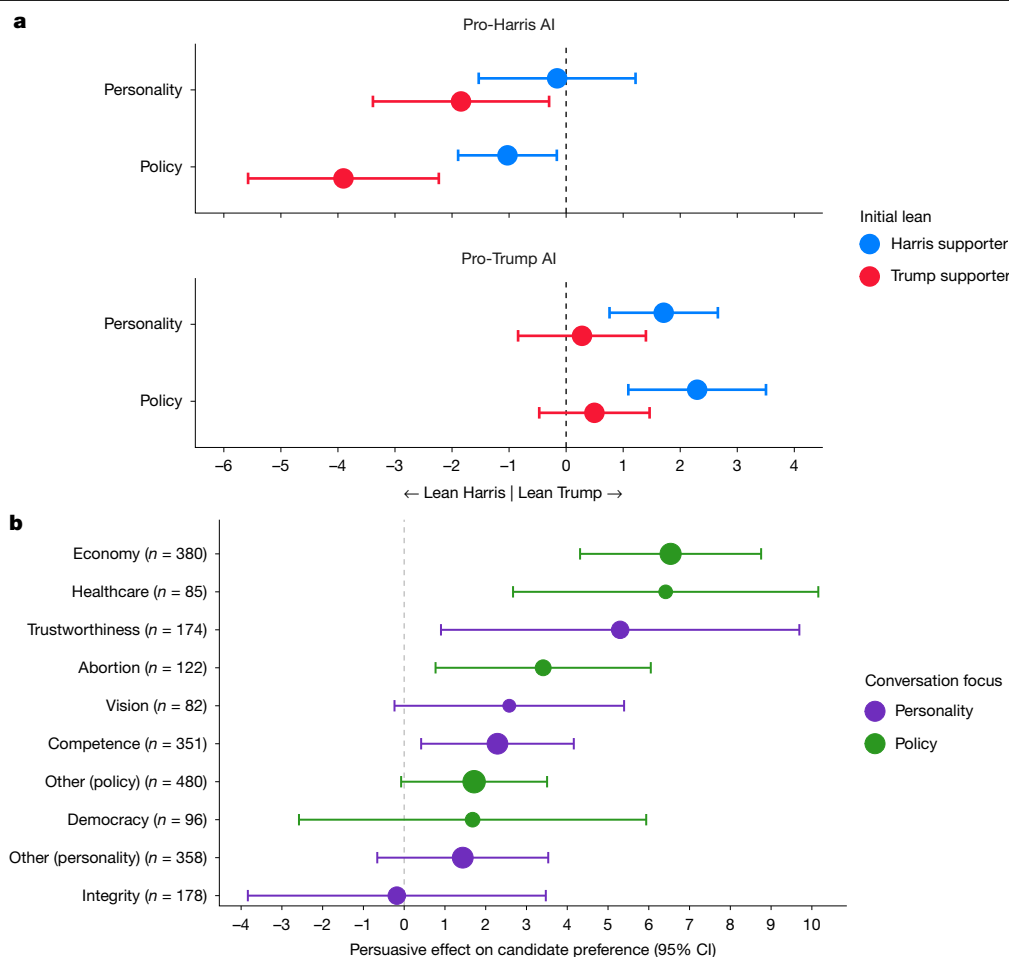


Fig. 1 | A conversation with an AI model changed participants' candidate preferences in the 2024 US presidential election. a, Pre-to-post change in preference for Harris versus Trump. The pro-Harris AI ($n = 1,255$) and pro-Trump AI ($n = 1,202$) treatments persuaded participants to prefer Harris and Trump more, respectively ($P < 0.001$). Both treatments were more effective at persuading people who initially opposed the candidate the AI was advocating for (P values < 0.001). Persuasion was also more effective when discussing policy ($P = 0.035$). Group means were compared using two-sided t -tests.

Error bars indicate 95% confidence intervals (CIs) of the mean. **b**, Persuasion effect (post-treatment difference between pro-Harris and pro-Trump conditions) shown separately by the policy issue/personal characteristic the participant selected as most important to them pre-treatment. Effects (group means) were estimated using linear regression with heteroscedasticity-robust s.e. controlling for pre-treatment attitude. Dot size is proportional to number of observations. Error bars indicate 95% confidence intervals of the mean.

important for them in deciding who to vote for in the upcoming election, or the candidate's personal characteristic that was most important to them (see Extended Data Fig. 1 for distributions), and then provide open-ended text responses describing why. Second, they were randomized to either the pro-Harris or pro-Trump persuasion condition, where they had a three-round back-and-forth conversation with the AI (median total duration 6.1 min). The AI model had two goals: (1) to increase support for the model's assigned candidate (pro-Harris AI or pro-Trump AI), and (2) to increase voting likelihood if the participant initially favoured the model's candidate, or decrease voting likelihood if the participant initially favoured the other candidate.

To achieve those goals, the model was instructed to be positive, respectful and fact-based; to use compelling arguments and analogies to illustrate its points and connect with its partner; to address concerns and counter arguments in a thoughtful manner and to begin the conversation by gently (re)acknowledging the partner's views. To facilitate personalization, the model was also provided with (pre-treatment) information about which candidate the participant intended to vote for and why, their 0–100 Harris versus Trump preference score and their voting likelihood.

After the chat, participants completed the outcome measures again. To assess the persistence of any effects induced by the treatment,

participants were re-contacted after more than a month (mean of 41.14 (s.d. = 6.85) days between initial survey and follow-up survey) and again completed the outcome measures ($n = 1,936$ completed the follow-up survey).

Effects of the human–AI dialogue

We focus on the effect of AI persuasion on candidate preference (Fig. 1a). To do so, we use linear regression predicting post-treatment candidate preference using persuasion condition (a pro-Harris versus pro-Trump dummy), conversation focus (a personal versus policy characteristic dummy, z-scored) and pre-treatment candidate preference (z-scored), along with all interactions. We also compute Q values that control false discovery rates using the Storey–Tibshirani method^{23,24}.

We found a significant main effect of persuasion condition (beta coefficient (b) = 2.88 (2.02, 3.74), $P < 0.001$, false discovery rate-adjusted P value (Q) < 0.001), which was qualified by a significant interaction with conversation focus ($b = 0.93$ (0.07, 1.79), $P = 0.035$, $Q = 0.022$), and a significant interaction with pre-treatment candidate preference ($b = 0.77$ (0.04, 1.51), $P = 0.038$, $Q = 0.022$). The interaction between conversation focus and pre-treatment candidate preference ($b = -0.12$ [–0.71, 0.47],

$P = 0.689$, $Q = 0.263$) and the three-way interaction ($b = -0.11$ [-0.84 , 0.62], $P = 0.768$, $Q = 0.285$) were not statistically significant.

Decomposing the interaction between persuasion condition and conversation focus shows that discussing policy was more effective than discussing personal characteristics. Relative to their initial attitude, participants in the pro-Harris condition became more pro-Harris after the conversation, especially when the conversation was about their top-priority policy issue (pre-to-post change: $b = 2.34$ (1.44 , 3.24), $P < 0.001$, $Q < 0.001$), and less so when it was their top-priority candidate personal characteristic (pre-to-post change: $b = 0.95$ (-0.08 , 1.97), $P = 0.071$, $Q = 0.037$). Similarly, participants in the pro-Trump condition became more pro-Trump after the conversation, with the effect being slightly larger when the conversation was about policy (pre-to-post change: $b = 1.51$ (0.71 , 2.31), $P < 0.001$, $Q < 0.001$) than personal characteristics (pre-to-post change: $b = 1.06$ (0.33 , 1.78), $P = 0.004$, $Q = 0.004$).

In a post hoc analysis, we find substantial heterogeneity in persuasion across the policy issue or personal characteristic participants selected as most important pre-treatment (random-effects meta-analytic Cochran's Q (d.f. = 9) = 22.02, $P = 0.009$; inconsistency index (I^2) = 59.68% of the variability in effect sizes across topics reflecting true differences rather than sampling error). Conversations about the economy, healthcare, and candidate trustworthiness produced the largest persuasion effects (Fig. 1b). See also Extended Data Fig. 7 for post-hoc analyses of heterogeneous effects based on participant characteristics.

To decompose the interaction between persuasion condition and pre-treatment candidate preference, we examined separately participants who initially favoured Harris ('Harris supporters') and those who initially favoured Trump ('Trump supporters'). The pro-Harris condition had a substantially larger effect on Trump supporters (policy: $b = 3.90$ (2.23 , 5.57), $P < 0.001$, $Q < 0.001$; personality: $b = 1.84$ (0.30 , 3.39), $P = 0.020$, $Q = 0.015$) than Harris supporters (policy: $b = 1.03$ (0.16 , 1.89), $P = 0.020$, $Q = 0.015$; personality: $b = 0.16$ (-1.22 , 1.53), $P = 0.822$, $Q = 0.297$). Conversely, the pro-Trump condition had a substantially larger effect on Harris supporters (policy: $b = 2.30$ (1.09 , 3.50), $P < 0.001$, $Q < 0.001$; personality: $b = 1.71$ (0.76 , 2.66), $P < 0.001$, $Q < 0.001$) than Trump supporters (policy: $b = 0.50$ (-0.47 , 1.46), $P = 0.313$, $Q = 0.138$; personality: $b = 0.28$ (-0.84 , 1.40), $P = 0.625$, $Q = 0.251$). Thus, both treatments were more effective on out-party participants than in-party participants, indicating that the treatment was successfully persuading people who were initially opposed to the candidate for whom the AI was advocating.

When examining effects on voting likelihood, each treatment increased the voting likelihood of in-party participants by slightly more than 2 points on the 0–100 scale, and had no significant effect on out-party participants (Supplementary Information section 2.1 and Extended Data Fig. 4a). We also conducted post hoc analyses examining treatment effects on what the participants reported they would do if the election happened today, and again found significant effects, such that after the conversation, participants in the pro-Harris condition were more likely to say they would vote for Harris (pre-to-post $b = 2.13$ (1.21 , 3.05), $P < 0.001$, $Q < 0.001$) and participants in the pro-Trump condition were more likely to say they would vote for Trump (pre-to-post $b = 1.41$ (0.57 , 2.26), $P = 0.001$, $Q = 0.003$) (Supplementary Information section 2.1 and Extended Data Fig. 4b). Thus, the treatment effects on candidate preference and voting likelihood in our pre-registered analyses translated into changes in actual vote intentions.

Finally, we examined persistence: to the extent that a participant's attitude changed immediately after the conversation, how much of that change remained when re-contacted more than a month later? Our pre-registered analyses predicted each participant's attitude change at follow-up (attitude at follow-up minus pre-treatment attitude) using their attitude change immediately after treatment (attitude immediately after treatment minus pre-treatment attitude). The coefficient on immediate effect indicates the fraction of the immediate effect

observed at follow-up (see Supplementary Information section 6.1 for simulations). We also included a control for pre-treatment attitudes to ensure that the relationship between the change scores is not a statistical artefact²⁵ (this control was not pre-registered, but results were virtually identical without it; Supplementary Information section 6.4). We found a significant relationship between immediate and delayed changes in candidate preference ($b = 0.34$ (0.23 , 0.44), $P < 0.001$, $Q < 0.001$), such that slightly more than a third of the original effect persisted more than a month later. We found similar results for voting likelihood and vote choice, and results were robust to making the very conservative assumption that any treatment effects decayed completely among the 16.05% of participants who did not return for follow-up ($b = 0.25$ (0.17 , 0.33), $P < 0.001$, $Q < 0.001$).

Persuading on a 2024 Massachusetts ballot measure

We then asked whether AI dialogues can be even more persuasive in the context of a state ballot measure concerning a relatively new issue that does not break cleanly along party lines: namely, a Massachusetts proposal to legalize psychedelic drugs²⁶. We recruited 501 Massachusetts residents in the week before the November 2024 election. Participants indicated pre-treatment attitudes towards drug legalization in general and support for the 2024 psychedelics legalization ballot measure specifically. As expected, attitudes were substantially less polarized than for the candidate choice experiment, with 7.75% of participants maximally opposing the measure and 24.5% of participants maximally supporting the measure (with support indicated using a 0–100 scale). Similar to the US Presidential election experiment, participants had a three- to five-round conversation with an AI model. To avoid ceiling and floor effects, randomization depended on initial support for psychedelics legalization. Those with moderate support (20–80) were assigned to either a pro-psychedelics AI persuading them to vote 'yes' or an anti-psychedelics AI persuading 'no'. Strong opponents (support less than 20) were randomized to the pro-psychedelics AI or a control condition (discussing preference for cats versus dogs), whereas strong supporters (support greater than 80) were randomized to the anti-psychedelics AI or the control. After the conversation, all participants again completed the pre-treatment measures.

Using linear regression to predict post-treatment ballot measure support using persuasion condition and pre-treatment support indicated a significant main effect of persuasion condition in each subgroup (Fig. 2). Among strong opponents, the pro-Psychedelic AI led to significantly more support post-treatment than the control ($b = 14.85$ (6.03 , 23.66), $P = 0.001$, $Q = 0.004$); among strong supporters, the anti-Psychedelic AI led to significantly less support post-treatment than the control ($b = -5.57$ (-8.87 , -2.26), $P = 0.001$, $Q = 0.004$); among non-extreme participants, the pro-Psychedelic AI led to significantly more support post-treatment than the anti-Psychedelic AI ($b = 22.73$ (17.00 , 28.46), $P < 0.001$, $Q < 0.001$). We see similar patterns of significant results on attitudes towards the legalization of psychedelics more broadly, but less effect on the other (secondary) measures (Supplementary Information section 4.2). These results indicate that human–AI dialogues may have an even bigger impact on state and local election outcomes than national elections.

Persuasion strategies used by the AI

To understand how the AI was persuading participants, we conducted post hoc analyses of the extent to which the AI model used different persuasion strategies. We combined strategies from the political persuasion academic literature and political persuasion practitioners (as synthesized by ref. 9) with strategies suggested by OpenAI's o1 AI model (following ref. 20), yielding a set of 27 unique strategies; see Supplementary Information section 8 for details. (We focused on US

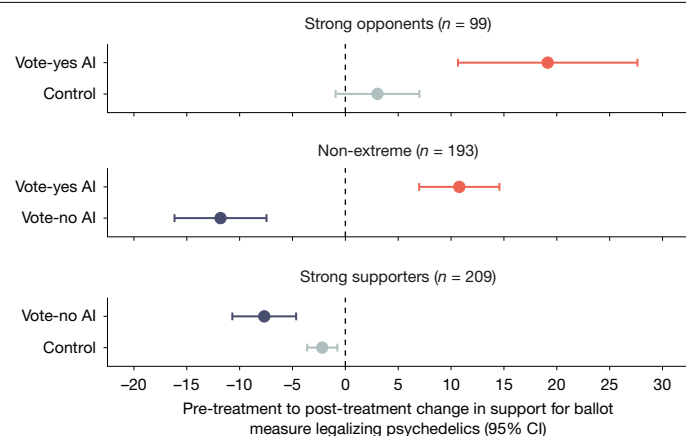


Fig. 2 | A conversation with an AI model changed Massachusetts residents' support for a ballot measure legalizing psychedelics. Pre-to-post change in support for legalizing psychedelics. Strong opponents of the measure (pre-treatment support less than 20; $n = 101$) were assigned randomly to interact with a pro-psychedelics AI or a control AI. Non-extreme participants (pre-treatment support between 20 and 80; $n = 204$) were assigned randomly to interact with an anti-psychedelics AI or pro-psychedelics AI. Strong proponents of the measure (pre-treatment support greater than 80; $n = 217$) were assigned randomly to interact with an anti-psychedelics AI or a control AI. Group means were compared using two-sided t -tests (P values ≤ 0.001). Error bars indicate 95% confidence intervals of the mean.

Presidential candidate preference persuasion, as the sample size in the ballot measure experiment was not large enough to yield stable estimates when predicting belief change with persuasion strategy use). To quantify strategy use, two LLMs (GPT-4o and Deepseek-V3) evaluated independently to what extent each strategy was used in each conversation using a three-level scale (0, 0.5, 1), and we averaged across the two evaluations.

The AI models used a diverse range of strategies (Fig. 3a). They were almost always polite and civil (average use level of 0.97 across conversations) and relied heavily on evidence and facts (0.64). They also tended to highlight optimistic outcomes while aligning messages with participants' core values (0.51), to encourage cognitive elaboration and critical reflection (0.45) and to engage in empathic listening (0.45). The AI models rarely used several strategies that are commonly discussed in the psychological and political science literature, such as making explicit calls to vote (0.11), storytelling (0.09), stimulating anger (0.05), using social influence strategies (0.04), name-calling (positive 0.13, negative 0.10) or using testimonials (positive 0.01, negative 0.04).

To gain insight into the effectiveness of each strategy in changing candidate preference, we conducted a post hoc analysis predicting pre-to-post attitude change using the prevalence of each of the 27 strategies in the dialogue, as well as claim accuracy (see below for details on estimation of accuracy) and experimental condition assignment. We used lasso (L1) regression with tenfold cross-validation to perform feature selection, which identified and retained 11 strategies (Fig. 3b). Personalizing the message to the participant and using evidence and facts were the strongest predictors of successful persuasion. Conversely, the more the AI model attempted to pre-emptively address potential objections, the less persuasion occurred—although this association may be explained, in part, by the AI model being more likely to pre-empt in conversations where the participant previously raised objections (and thus was less likely to change their minds at baseline). One particularly interesting lack of predictive power was for claim accuracy, which was excluded by the model, indicating there were no substantial persuasive gains associated with attempting to mislead.

Testing mechanisms in Canada and Poland

To shed further light on the mechanisms underlying AI political persuasion, we conducted two experiments in the context of high-salience national elections in Canada and Poland.

In one experiment, conducted the week before 28 April 2025 Canadian federal election experiment, 1,530 Canadians were randomized to talk to AI models that advocated for either the Liberal Party leader Mark Carney or the Conservative Party leader Pierre Poilievre. We also randomized whether the AI model was given similar instructions to the models in the US experiment (baseline condition) or was explicitly instructed to not use facts and evidence when making its arguments (no-facts condition; the models used the facts and evidence strategy much less and instead relied on analogies and general arguments; Supplementary Information section 8.1 and Extended Data Fig. 8). The experimental design was otherwise the same as the US election experiment, except that all participants indicated and discussed the policy that was most important to them (that is, the personal characteristic condition was eliminated; see Extended Data Fig. 2 for distributions); and we removed the part of the prompt that instructed the AI to reduce vote likelihood among counter-partisans.

In the baseline condition (Fig. 4a), we replicate our previous findings, with a significant difference between the pro-Carney and pro-Poilievre treatments ($b = 10.05$ (7.52, 12.57), $P < 0.001$, $Q < 0.001$). Critically, however, this persuasion effect was substantially smaller in the no-facts condition compared with the baseline ($b = -5.88$ (-9.31, -2.46), $P = 0.001$, $Q = 0.003$) (Fig. 4a and Supplementary Information section 4.3). Thus, at baseline, the persuasive effect was almost three times larger in the Canadian federal election than the effect observed in the US experiment, but depriving the AI of the ability to use facts and evidence reduced the effect by more than half.

In another experiment, conducted in the 2 weeks before the 18 May 2025 Polish presidential election, 2,118 Poles were randomized to talk to AI models that advocated for either the centrist-liberal Civic Coalition's candidate Rafał Trzaskowski or the right-wing Law and Justice party's preferred candidate Karol Nawrocki (Extended Data Fig. 3). We randomized across four strategies: the same baseline and no-facts conditions as in the Canada experiment, as well as two more conditions. In the non-personalized condition, the model was explicitly instructed to not personalize its responses and instead deliver broad, universal arguments (indeed, the non-personalized condition used the personalization strategy relatively infrequently; Supplementary Information section 8.2 and Extended Data Fig. 9). In the non-specific condition, we simply gave the model the directive to persuade with no specific instructions about how to do so. Otherwise, the experimental design was the same as the Canadian experiment.

In the baseline condition (Fig. 4b), we again replicated our previous findings with a significant difference between the pro-Trzaskowski versus pro-Nawrocki treatments ($b = 9.94$ (6.68, 13.20), $P < 0.001$, $Q < 0.001$). We also replicate the finding from the Canadian experiment, whereby the persuasive effect was significantly smaller in the no-facts condition compared with the baseline ($b = -7.74$ [-11.77, -3.71], $P < 0.001$, $Q < 0.001$). Conversely, we found no significant difference between the baseline and either the non-personalized strategy ($b = -1.58$ (-6.05, 2.88); $P = 0.487$, $Q = 0.064$, Bayes factor favouring null: 25.34) or the non-specific strategy ($b = -1.82$ (-6.31, 2.68), $P = 0.428$, $Q = 0.064$; Bayes factor favouring null: 23.80) (Fig. 4b and Supplementary Information section 4.4). Thus, as in the Canadian experiment, the baseline persuasive effect in Poland was almost three times larger than in the US experiment; and depriving the AI of the ability to use facts and evidence reduced the treatment effect by 78%. The lack of difference between the baseline and the non-personalized condition, conversely, indicates that although facts and evidence are a key mechanism for AI persuasion, personalization is not (at least in the Polish context). Similarly, it seems that simply

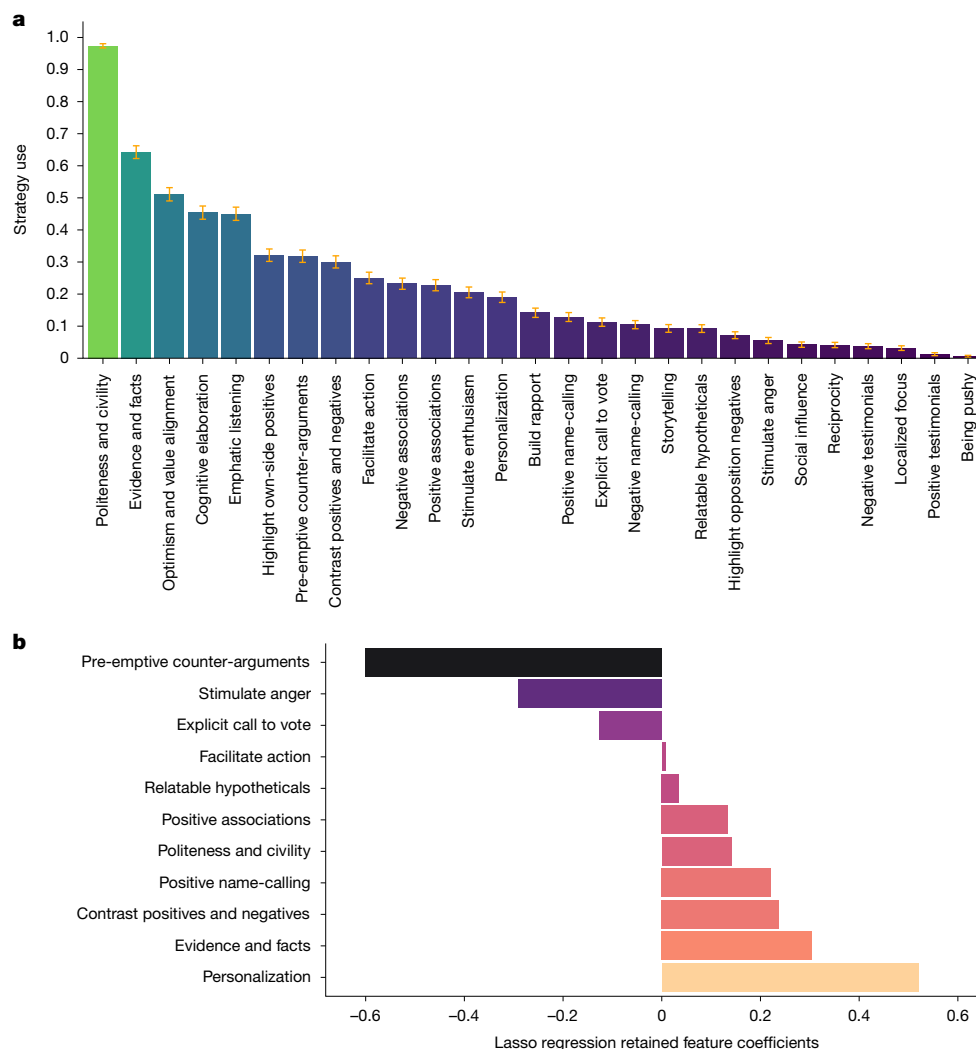


Fig. 3 | Strategies used by the AI model in the US presidential election experiment. a, Average use of the 27 strategies. Two AI models rated the use of each strategy using a three-level scale (0, 0.5, 1) and those two ratings were averaged. Error bars indicate 95% confidence intervals of the mean. **b**, Lasso (L1) regression was used to predict persuasive effects (pre-to-post change in

candidate preference) and select important strategies. The coefficients for the 11 strategies retained by the lasso model are shown. Positive (negative) coefficients indicate that use of a particular strategy is associated with greater (lesser) persuasive effects.

asking the model to persuade was not meaningfully less effective than the particular instructions we gave our AI model (and we find little difference in the strategies used in the baseline versus non-specific conditions; Supplementary Information Section 8.1). See Extended Data Figs. 5 and 6 for effects on voting likelihood and vote choice.

Finally, we also randomized which AI model we used (GPT4.1, DeepSeek-V3-0324, and Llama-4-Maverick in the Canadian experiment; GPT4.1 and DeepSeek-V3-0324 in the Polish experiment). We found no significant difference in persuasiveness across models. Thus, our results are not specific to one particular model (Supplementary Information sections 4.3 and 4.4).

Accuracy of AI model claims

We also shed light on the extent to which frontier models persuade people by providing accurate information versus misinformation (despite us explicitly instructing the models to be fact-based). We conducted post hoc fact-checking using Perplexity AI's online LLM; we validated this procedure using agreement with professional fact-checkers and showed that its ratings were not biased politically using agreement

with politically balanced groups of laypeople (Supplementary Information section 7.3).

All 8,134 AI model statements were rated on a scale from 0 (completely inaccurate) to 100 (completely accurate). Overall, the statements made by the AI models were mostly accurate (median = 90, median absolute deviation = 14.83). In a post hoc analysis comparing accuracy scores across conditions, however, we find stark differences (Fig. 5a). The pro-Trump AI made substantially more inaccurate statements compared with the pro-Harris AI ($b = -22.46$ ($-23.77, -21.15$), $P < 0.001$). For example, the AI argued Trump's competence was shown by tax reforms and job growth; Perplexity rated this inaccurate for relying on selective facts and ignoring that growth continued from Obama's presidency. It also claimed Harris's consistency was shown by commitments to healthcare, racial justice and immigration; Perplexity rated this inaccurate because she had shifted stances on these issues. See Supplementary Information section 7.1 and Supplementary Table 40 for more examples.

We observed the same political asymmetry in arguments generated using 12 different models (Supplementary Information section 7.4 and Extended Data Fig. 10) and in the Canadian and Polish experiments ($P < 0.002$; Fig. 5b,c and Supplementary Information section 7.6).

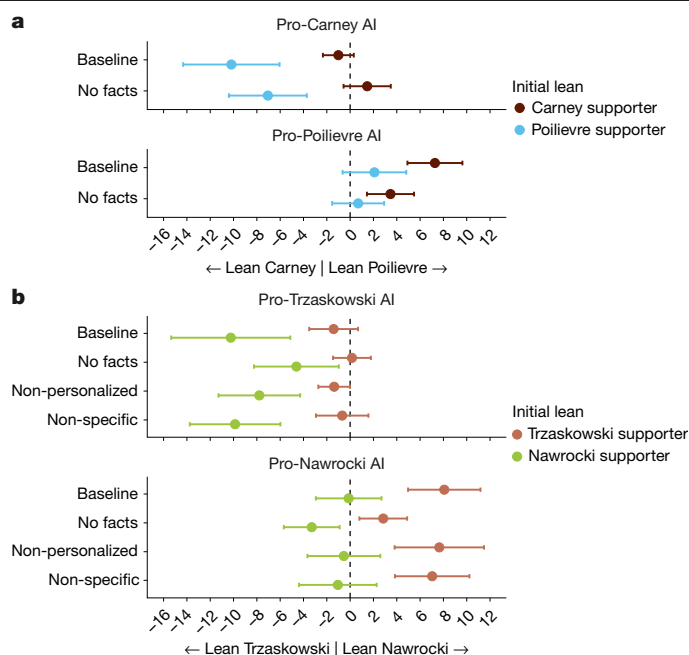


Fig. 4 | A conversation with an AI model changed participants' candidate preferences in the 2025 Canadian and Polish elections. Shown are the pre-to-post changes for one candidate versus the other. **a**, The pro-Carney AI ($n = 844$) and pro-Poillievre AI ($n = 850$) persuaded participants to prefer Carney and Poillievre more, respectively ($P < 0.001$). Persuasion effects were larger for out-party participants than in-party participants, and smaller when the AI was prompted to not use facts ($P = 0.001$). **b**, The pro-Trzaskowski AI ($n = 1,196$) and pro-Nawrocki AI ($n = 1,184$) persuade participants to prefer Trzaskowski and Nawrocki more, respectively ($P < 0.001$). The persuasion effects were larger for out-party participants, and smaller when the AI was prompted to not use facts ($P < 0.001$). When the AI did not personalize or were not given instructions on how to persuade, it was as effective in persuading as the AI in the baseline condition (P values ≥ 0.428 , Bayes factor favouring null > 23). Group means were compared using two-sided t -tests. Error bars indicate 95% confidence intervals of the mean.

Thus, across three experiments, countries and numerous different AI models, the pro-right-leaning candidate (Trump, Poillievre and Nawrocki) AI models made more inaccurate claims than the pro-left-leaning candidate (Harris, Carney and Trzaskowski) AI models.

Discussion

There is widespread concern about the potential for generative AI to influence electoral politics. Our results unambiguously demonstrate across three different countries, with different electoral systems, that dialogues with language models can meaningfully change voter attitudes and voting intentions. This observation has implications for the future of political persuasion, political advertising and (more broadly) democracy.

To contextualize the magnitude of these persuasion effects, we compare them with large-scale survey experiments evaluating traditional political advertisements^{8,9} or static persuasive text^{27,28}. Although there are important differences between these studies and ours, they are similar in fundamental ways: the treatments were delivered in survey experiments to 'captive audience' participants (that is, participants did not have to opt into receiving the treatment by opening their door in response to a canvasser's knock), and attitudes were measured in-survey using self-reports administered immediately after the treatment. Starting with the US experiment, we focus on the policy conditions as they yielded larger effect sizes (and are the approach campaigns are most likely to deploy). We find a 3.90-point change among

Trump supporters in the pro-Harris policy condition, compared with 0.98 point and 0.89 point treatment effects among Republicans seeing pro-Democrat advertisements in 2016 (ref. 8) and 2020 (ref. 9), respectively; and a 2.30-point change among Harris supporters in the pro-Trump policy condition, compared with a -0.65 -point treatment effect among Democrats seeing pro-Republican advertisements in 2016 (ref. 8). Contextualizing the effectiveness of the AI treatments in our ballot measure experiment is more difficult as treatment effects vary substantially across issues, and previous work has not examined this particular ballot measure. Be that as it may, we compare our observed average change in belief of 9.6 points for the anti-psychedelic condition and 13.6 points for the pro-psychedelics condition with previous work testing professionally produced left-leaning video advertisements across 36 issues, which found an average treatment effect of 4.5 points, and 2.25 points for text-only versions of the treatments²⁸; and a large experiment testing text-based messages taking opposition positions on 24 issues, which found an average treatment effect of 5.5 points²⁷. Analogous benchmarking studies were not available for the Canadian and Polish election contexts.

Taken together, these comparisons indicate that the AI dialogues were meaningfully more persuasive than traditional persuasion approaches. Furthermore, given that we find some evidence that the model's persuasiveness varied substantially depending on which policy or personal characteristic was discussed (Fig. 1b), it is possible that even larger effects would have been observed if participants had been assigned to discuss one of the topics with particularly large effects, rather than discussing the topic they selected as most important to them.

Dominant theories of politically motivated reasoning^{29–32} argue that political motivations blind people to facts and evidence that challenge their existing political positions, and thus that presenting uncongenial facts and evidence will have little effect. Instead, these theories imply that reliance on other, more psychologically oriented, approaches to persuasion will be more effective³³. Yet our data indicate that the AI models are persuading potential voters by politely providing relevant facts and evidence, rather than by being skilled manipulators who leverage sophisticated psychological persuasion strategies such as social influence^{34–36}, storytelling^{33,37}, reciprocity³⁵, testimonials^{38,39} or stimulating anger^{40,41}. These findings are consistent with the AI operating with the central, rather than peripheral, route to persuasion⁴². Furthermore, we found that AI persuasion was particularly effective for out-party members (for example, it was more persuasive when arguing for Harris among Trump supporters than among Harris supporters). Our results therefore demonstrate the power of uncongenial facts and evidence to change political attitudes—a finding that is starkly inconsistent with the predictions of theories of politically motivated reasoning.

Conversely, our results are consistent with recent arguments that evidence-based reasoning is more pervasive than previously thought⁴³. For example, recent work shows that evidence-based dialogues with AI were surprisingly effective at decreasing belief even among apparently entrenched conspiracy believers²⁰, specifically because of the facts and evidence presented by the AI²¹. It is noteworthy that the effects reported for conspiracy debunking are similar in magnitude to the effects we find for the Canadian and Polish elections and the psychedelics ballot measure, but that the effects we find for the US election are substantially smaller. Potential explanations for the smaller effects in the US election compared with the Canadian and Polish elections include the possibility of participants having received more pre-treatment exposure to information about Trump (and, to a lesser extent, Harris) than the Canadian or Polish candidates, the difference between two-party and multi-party elections, and potential mixed messaging from the AI in the US experiments, where it was instructed to both persuade and demotivate out-partisans.

Another interesting theoretical implication involves the role of personalization. Whereas a large body of previous research has indicated

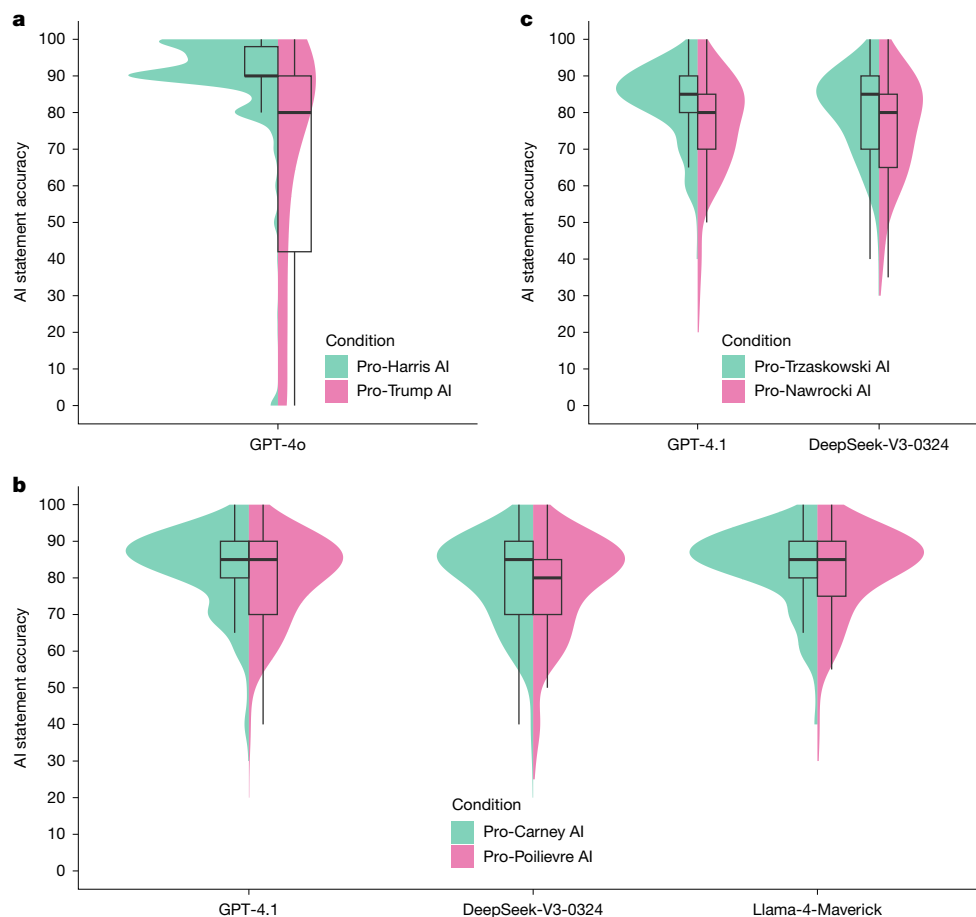


Fig. 5 | AI models supporting the more right-leaning candidates made more inaccurate claims than the AI models supporting the more left-leaning candidates. Shown are distributions of accuracy ratings for different randomly assigned conditions and models. **a**, In the US experiment, the pro-Trump AI's claims ($k = 4,019$) were less accurate ($P < 0.001$) than the pro-Harris AI's claims ($k = 4,115$). **b**, In the Canadian experiment (baseline condition), the pro-Poillievre AI's claims ($k = 1,552$) were less accurate ($P = 0.001$) than the pro-Carney AI's

claims ($k = 1,426$). **c**, In the Polish experiment (baseline condition), the pro-Nawrocki AI's claims ($k = 970$) were less accurate ($P < 0.001$) than the pro-Trzaskowski AI's claims ($k = 1,067$). The accuracy ratings were determined by Perplexity, a language model that searches the internet for real-time information and that we find produces high agreement with professional fact-checkers. Boxplots show the median, interquartile range and whiskers (extending to 1.5 times the interquartile range).

the importance of personalizing persuasive messages to align with participants' values, identity or personal disposition^{35,44–46}, we did not find clear evidence of large returns to personalization. Responding to the person's specific beliefs and concerns was predictive of model persuasiveness in post hoc analyses of the US experiment, but an experimental manipulation in the Polish study had no significant effect. These results demonstrate that personalization is not a necessary ingredient for large persuasive effects, even in high-salience contexts such as presidential elections. This resonates with recent findings on LLM-based policy issue persuasion, which variously found no significant effect⁴⁷ or a precisely estimated very small effect¹⁹ of personalization.

A particularly striking result in our data is the asymmetry in the accuracy of the facts and evidence being provided by the bots, wherein AI models advocating for candidates on the political right (Trump, Poillievre and Nawrocki) made more inaccurate statements than AI models advocating for candidates on the political left (Harris, Carney and Trzaskowski). This asymmetry is not easily explained by liberal bias in the accuracy judgments, which were generated by the Perplexity AI model: a politically balanced set of laypeople consistently agreed with Perplexity's evaluations. Past work has shown that, in the USA, misinformation is shared more by those on the political right than on the left^{48–52}—even when information quality is assessed by politically balanced crowds⁵³ or Community Notes on X⁵⁴. It is surprising and societally important that generative AI seems to replicate these

patterns found in the broader information environment despite the guardrails put in place to ensure accuracy. It remains unclear, however, whether dishonest persuasion is more effective, as we find no association between accuracy and persuasiveness (although other work has found that more persuasive models are less accurate¹⁹). Future work should investigate this issue more directly by experimentally manipulating the accuracy of information provided.

We undertook this research because of the societal gains from a threat assessment of AI interactions in political discourse, but were mindful of the potential risks. To minimize the potential real-world impacts, we informed participants they were interacting with an AI model and provided a debriefing that informed them the AI had been explicitly instructed to persuade them to prefer one candidate, and we reminded them of AI limitations and provided resources for informed voting decisions. We also randomized the direction of persuasion so that any unintended longer-term effects should cancel out, minimizing any overall impact on the elections. Debriefing participants before the election means we could examine only self-report measures, but our results are unlikely to be explained by demand effects. Previous work shows little evidence of demand effects in similar contexts⁵⁵, and our no-facts condition imposed similar demands—providing clear evidence of a persuasive effect above and beyond demand effects.

In sum, we have demonstrated that human–AI dialogues can meaningfully impact voter attitudes across three disparate countries. Getting voters—particularly opposition supporters, who responded most to the treatment—to choose to engage with an AI chatbot will present a major challenge to deployment in the field. Furthermore, at least in the US presidential election context, the magnitude of the effects we observed was not that much larger than effects observed from traditional political advertisements, and people who sign up for survey panels and chose to participate in studies involving AI may show larger effects than those who do not opt in to such studies. Thus, it remains to be seen how effective a technology such as this will be if actually deployed by political campaigns. Be that as it may, it seems highly likely that AI-based approaches to persuasion will play an important role in future elections—with potentially profound consequences for democracy.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-09771-9>.

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Methods

US presidential candidate experiment

Inclusion and ethics. The study was conducted from 22 August 2024 to 5 September 2024, and was deemed exempt by the Massachusetts Institute of Technology (MIT) Committee on the Use of Humans as Experimental Subjects (protocol E-5990). We obtained informed consent from participants. The study did not involve deception, and participants were told they would be conversing with an AI in advance. After the study was completed, participants were debriefed (Supplementary Information section 10.1.1). The study was pre-registered (<https://aspredicted.org/s4kr-9n8j.pdf>). Non-pre-registered analyses are labelled post hoc. In line with our pre-registration, participants who failed a pre-treatment free-text attention/effect filter were not allowed to participate.

Participants. We pre-registered a target sample of 2,400 US participants, recruiting from CloudResearch Connect and Prolific (Supplementary Information section 10.1). In total, 2,457 participants started the survey, and 2,306 (1,063 Democrats, 874 Republicans, 369 Independents) completed the survey; see Supplementary Fig. 1 for demographic distributions.

Materials and procedure. Participants assigned to indicate the policy issue that was most important to them in deciding who to vote for in the upcoming US 2024 Presidential Election selected from a list of 19 policy issues. Those assigned to indicate the candidate's personal characteristic the most important to them selected from a list of 19 characteristics. Before conversing with the AI, participants indicated their preference for Harris versus Trump using a 0–100 scale (0, strong Harris; 100, strong Trump) and their likelihood of voting using a 0–100 scale (0, definitely will not vote; 100, definitely will vote). We also examined a post hoc categorical vote choice question ('Imagine if the 2024 Presidential Election were today, who would you vote for, if anyone?'; options: Kamala Harris, Donald Trump, Third-party candidate (please specify), and so on). Participants also completed several items about their attitudes (for example, trust towards AI technologies) and demographic characteristics (for example, age, gender, education) (Supplementary Information section 10.1.1).

Participants were then assigned randomly to condition using a two (conversation focus: policy versus personal characteristic) by two (pro-Harris versus pro-Trump AI) between-subjects design. After selecting their top-priority policy/characteristic, participants explained their choice (Supplementary Information section 10.1.1). Responses to these two open-ended questions were fed to the AI prompt (Supplementary Information section 9.1). Participants were also assigned randomly to either the pro-Harris or pro-Trump persuasion condition, in which the AI was prompted to explain why the assigned candidate is better. The AI was also prompted to increase voting likelihood for participants who initially supported the assigned candidate (Harris supporter assigned to pro-Harris condition or Trump supporter assigned to pro-Trump condition) and to decrease voting likelihood for participants who initially supported the opposing candidate (Harris supporter assigned to pro-Trump condition or Trump supporter assigned to pro-Harris condition). For prompting the AI, participants with scores 50 or below on candidate preference were classified as Harris supporters, whereas those above 50 were classified as Trump supporters—we adopted the same classification convention in our analyses.

Participants were told that they would be conversing with an AI about some of the topics and ideas previously mentioned in the survey. They then engaged in a three-round conversation with the AI (prompts are shown in Supplementary Information section 9.1). To make the AI-generated content as factually accurate, up-to-date, conversational and engaging as possible, we developed a multi-agent system deployed by the Vegapunk platform: <https://www.vegapunkdoc.dev> (Supplementary Information section 10.1.1).

After the conversation with the AI, participants completed a subset of items they completed pre-treatment: preference for Harris versus Trump, voting likelihood, categorical vote choice, feelings towards Harris, feelings towards Trump and trust towards AI technologies such as ChatGPT (Supplementary Information section 10.1.1).

Analysis. The pre-registered outcome variables are candidate preference and voting likelihood; categorical vote choice was post hoc. We fit linear regression models with heteroscedasticity-robust (HC1) s.e. values to estimate treatment effects. We predict post-treatment candidate preference using persuasion condition (pro-Harris (coded as 0) versus pro-Trump (coded as 1)), conversation focus (personal (coded as 0) versus policy characteristic (coded as 1), z-scored), and pre-treatment candidate preference (z-scored), along with all interactions (see Supplementary Information section 4.1 for full regression tables). (We pre-registered using bootstrapping to compare these changes, but omit these analyses because they are redundant with the main regression models.) We test whether treatment changes voting likelihood using two separate models (Supplementary Information section 10.1.1). We follow up using one-sample *t*-tests comparing pre- and post-treatment values. We also control for the false discovery rate across multiple comparisons using the Storey–Tibshirani method^{23,24} (Supplementary Information section 10.1.1).

Persuasion strategies. We perform post hoc analyses to identify 27 strategies (Fig. 3a and Supplementary Information section 8) from the political persuasion academic literature and practitioners (as synthesized by ref. 9) or suggestions by an AI model (following ref. 20). Separately for each conversation and strategy, two language models independently evaluating the extent the strategy were used, and we took the mean across the evaluations of the two models. We then performed lasso regression to identify important strategies^{56,57} (Supplementary Information section 10.1.1).

US experiment follow-up

Participants. More than a month after participants completed the main study, we re-contacted them to invite them to participate in a follow-up survey (pre-registration: <https://aspredicted.org/vvcf-bx5n.pdf>). Among those who completed the main study, 1,936 (83.95%) completed this re-contact study (Supplementary Information section 10.1), which ran from 4 October to 2 November 2024. On average, participants completed the follow-up survey after 41 days (s.d. = 6.85; range, 30–63 days).

Materials and procedure. Participants completed the main measures reported in the main study: candidate preference, voting likelihood and categorical vote choice. They also indicated how closely they have been following news about candidates for the US 2024 Presidential Election (0, not at all; 100, extremely) and were asked to describe anything they could recall about their previous conversation with the AI.

Analysis. We evaluated the persistence of the treatment effects on outcome variables as follows: let t_0 , t_1 , and t_2 be an outcome variable measured pre-treatment in the main study, immediately post-treatment in the main study and the re-contact study, respectively; then let d_2 and d_1 be the difference between t_2 and t_0 ($t_2 - t_0$), and the difference between t_1 and t_0 ($t_1 - t_0$), respectively. For each outcome, we fit a linear regression predicting d_2 using d_1 while controlling for t_0 . The coefficient on d_1 indicates the fraction of the immediate treatment effect still observed at re-contact (see Supplementary Information section 6.1 for simulations supporting the interpretation of this coefficient as capturing the persistence relationship). We also fit a mediation model to test for an indirect effect (Supplementary Information sections 6.3 and 10.2). As robustness checks, we conservatively assume that treatment effects decayed completely among the 16.05% of participants who did not

return to complete this re-contact survey study (that is, replace all missing t_2 values with that participant's t_0 value).

Ballot measure support experiment

Inclusion and ethics. The study was conducted from 30 October 2024 to 5 November 2024 (pre-registration: <https://aspredicted.org/q87m-j56v.pdf>), and was deemed exempt by the MIT Committee on the Use of Humans as Experimental Subjects (protocol E-5990). We obtained informed consent from participants. The study did not involve deception, and participants were told they would be conversing with an AI in advance. After the study was completed, all participants were debriefed. Analyses that were not pre-registered are labelled post hoc, and deviations from pre-registrations are reported.

Participants. We pre-registered a target sample of 500 Massachusetts residents. We initially pre-registered recruiting 250 Democrats and 250 Republicans but, given the tight timeline for recruitment and the limited pool of Massachusetts residents available, we did not implement partisanship quotas. We recruited participants from both Prolific and CloudResearch Connect. In total, 522 participants began the survey (Supplementary Information section 10.3). A total of 21 participants dropped out during the treatment, and the dropout rate did not differ significantly across conditions, $\chi^2(2) = 0.49, P = 0.784$ (Supplementary Information section 10.3 and Supplementary Fig. 20). This results in a final set of 501 participants (291 Democrats, 85 Republicans, 125 Independents; mean age = 38.73 years (s.d. = 13.74); 266 female, 213 male and 22 who selected another gender option); see Supplementary Fig. 2 for distributions of pre-treatment covariates.

Materials and procedure. After completing demographics, a question on trust in generative AI, and political preference questions, participants indicated whether they themselves, or someone they knew, had experimented with psychedelics in the past; whether they believe drug legislation should be a state versus federal responsibility; their attitude towards total illicit drug legalization; and their attitudes towards legalization of cannabis, psychedelic and synthetic drugs (all outcomes were measured using 0–100 scales). Next, we explained MA Ballot Question 4, asked participants for their position on this measure using a 0–100 scale from ‘Strongly No’ to ‘Strongly Yes’, and had them explain their attitudes (Supplementary Information section 10.3.1).

On the basis of their position on the ballot measure, participants were then assigned to one of three conditions. Participants with an initial attitude between 20 and 80 were assigned randomly to be persuaded to vote yes or no on Ballot Question 4. Those below 20 were assigned randomly to be persuaded to vote yes or to a control. Those above 80 were assigned randomly to be persuaded to vote no or to a control. Analyses were conducted separately on these three subgroups (initial likelihood to vote Yes less than 20—‘strong opponents’; initial likelihood between 20 and 80—‘non-extreme’; initial likelihood more than 80—‘strong supporters’) (Supplementary Information section 10.3).

Participants then engaged in a three- to five-round back-and-forth conversation with the AI (OpenAI GPT-4o; prompts are shown in Supplementary Information section 9.2). Participants in the vote Yes and vote No conditions interacted with AI prompted to persuade the participant to vote Yes or vote No, respectively. The prompt included the participants’ text response to the question about their attitudes towards the ballot measure. Participants assigned to the control group engaged in a conversation with an AI that was prompted to persuade the participant that cats are better pets than dogs (due to a programming error, participants in the control were informed they were going to talk to the AI about psychedelic drugs, but then once the conversation began the AI immediately opened by discussing cats versus dogs and did not bring up psychedelics).

After the conversation with the AI, participants again completed a subset of items they had completed pre-treatment: trust in AI, likelihood

to vote ‘Yes’ on the ballot question, whether they think drug legalization should be a state or federal responsibility, their attitude towards total illicit drug legalization and their separate attitudes towards cannabis, psychedelic and synthetic drug legalization. They also completed three extra items about their experience with the AI (‘How well did you feel the AI in this conversation understood your perspective?’, ‘How respectful did you feel the AI was in this conversation?’ and ‘The AI made points that I hadn’t thought of before’).

Analysis. Our pre-registered main outcome variable is support for the psychedelic legalization measure. Our pre-registered secondary outcome variable is belief that psychedelics should be legal. We also pre-registered spill-over effects on the extent to which participants believe drugs should be illegal, believe cannabis should be legal, believe synthetic drugs should be legal and believe drug legalization should be a state versus federal issue. For each outcome, we fit linear regression models with HC1 s.e. values predicting the post-treatment value using a condition dummy and a control for pre-treatment value, separately for strong opponents, non-extreme participants and strong supporters (see Supplementary Information section 4.2 for full regression tables). We also control for the false discovery rate across multiple comparisons using the Storey–Tibshirani method^{23,24} (Supplementary Information section 10.3.1).

Canadian federal election experiment

Inclusion and ethics. The study was conducted from 23 April 2025 to 28 April 2025, and was deemed exempt by the MIT Committee on the Use of Humans as Experimental Subjects (protocol E-6748). We obtained informed consent from participants. The study did not involve deception, and participants were told they would be conversing with an AI in advance. The study was pre-registered (<https://aspredicted.org/bft5-fc9j.pdf>). Any non-pre-registered analyses are labelled post hoc, and deviations from pre-registrations are reported.

Participants. We pre-registered a target sample of 4,000 Canadians, recruiting from four platforms (CloudResearch Connect, Lucid, Prolific, PureSpectrum) in the 5 days leading up to the Canadian elections on 28 April 2025. In total, 1,694 participants began the survey, and 1,530 completed the survey (850 Liberals, 490 Conservatives, 211 neither Liberal nor Conservative; Supplementary Information section 10.4). This sample (mean age, 41.41 years (s.d. = 16.67)) included 718 female participants, 797 male participants and 15 who selected another gender option; see Supplementary Fig. 3 for distributions of pre-treatment covariates.

Materials and procedure. Participants indicated the policy issue that was most important to them in deciding who to vote for in the upcoming Canadian federal election. They selected from a list of 17 policy issues. The procedure is similar to that of the US presidential election experiment, but with the following changes. Canada operates under a parliamentary system where voters elect Members of Parliament to represent their local ridings; the party leader whose party wins the most seats becomes the Prime Minister. Thus, the categorical vote choice question was rephrased accordingly; Supplementary Information section 10.4.1. Whether a participant was a Carney or Poilievre supporter was determined on the basis of their preference for Carney versus Poilievre (measured using a 0 (strong Carney) to 100 (strong Poilievre) scale).

Participants were assigned randomly to a condition using a two (persuasion condition: pro-Carney versus pro-Poilievre AI) by two (strategy: baseline versus no facts) by three (model: GPT4.1 versus DeepSeek-V3-0324 versus Llama-4-Maverick) between-subjects design; Supplementary Information section 10.4. Unlike the US experiment, the AI was prompted to increase voting likelihood for participants who initially supported the assigned candidate and to not change voting

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likelihood for participants who initially supported the opposing candidate. Otherwise, in the baseline strategy condition, the prompt was identical to the one in the US experiment; in the no-facts strategy condition, the AI model was prompted to never use facts and reason-based persuasion strategies (exact prompts are shown in Supplementary Information section 9.3).

Analysis. The pre-registered primary outcome variable is candidate preference. The secondary outcomes are voting likelihood and categorical vote choice. These variables were measured before and after treatment. The modelling approach is similar to the one used in the US experiment. We also examined whether the treatment effects vary by the AI used, restricting to participants in the baseline strategy condition. See Supplementary Information section 4.3 for full regression tables. We also control for the false discovery rate across multiple comparisons using the Storey–Tibshirani method^{23,24} (Supplementary Information section 10.4.1).

Polish presidential election experiment

Inclusion and ethics. The study was conducted from 7 May 2025 to 16 May 2025, and was deemed exempt by the MIT Committee on the Use of Humans as Experimental Subjects (protocol E-6757). We obtained informed consent from participants. The study did not involve deception, and participants were told they would be conversing with an AI in advance. After the study was completed, all participants received a debriefing similar to that in the US experiment. The study was pre-registered (<https://aspredicted.org/42d9-8trn.pdf>). Non-pre-registered analyses are labelled post hoc, and deviations from pre-registrations are reported.

Participants. We pre-registered a target sample of 4,000 Poles recruited from Lucid. In total, 2,380 participants began the survey, and 2,118 completed the survey (mean age, 39.41 years (s.d. = 20.24); 904 female voters, 1,196 male voters and 18 selected another gender option); see Supplementary Information section 10.5 for details and Supplementary Fig. 4 for distributions of pre-treatment covariates.

Materials and procedure. Participants indicated the policy issue that was most important to them in deciding who to vote for in the Polish 2025 presidential election by selecting from a list of 18 policy issues (Supplementary Information section 10.5.1). The procedure is similar to that of the Canadian experiment, but with the following changes. The two candidates are Rafał Trzaskowski, a centre-left candidate (member of the Civic Platform Party), and Karol Nawrocki, a right-wing candidate (supported by the Law and Justice Party)—they were the top candidates in the first round of Poland presidential elections, according to the polls.

Participants were assigned randomly to a condition using a two (persuasion condition: pro-Trzaskowski versus pro-Nawrocki AI) by four (strategy: baseline versus no facts versus non-personalized versus non-specific) by two (model: GPT4.1 versus DeepSeek-V3-0324) between-subjects design; Supplementary Information section 10.5. In addition to the two strategies (baseline and no facts) used in the Canadian experiment, we include two more strategies: a non-personalized strategy where the AI was prompted to present generic persuasive messages without personalization, and a non-specific strategy where the AI was not given specific instructions on how to persuade (prompts are shown in Supplementary Information section 9.4).

Analysis. The pre-registered primary outcome variable is candidate preference (0, strong Trzaskowski; 100, strong Nawrocki). The secondary outcomes are voting likelihood and categorical vote choice. These variables were measured before and after treatment. The modelling approach is similar to the one used in the US and Canadian experiments; see Supplementary Information section 4.4 for full regression tables. We also control for the false discovery rate across multiple comparisons using the Storey–Tibshirani method^{23,24} (Supplementary Information section 10.5.1).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Due to the sensitive nature of this research—namely, the dual-use potential associated with the infrastructure and data used for political persuasion—we do not publicly share raw conversation transcripts or the bespoke LLM code. Release of these materials could facilitate misuse, such as the development or fine-tuning of models for unethical persuasion. These materials may be made available from the authors upon reasonable request through Harvard Dataverse at <https://doi.org/10.7910/DVN/DODEXZ> (ref. 58). Requests will be evaluated on an individual basis, considering the researcher's credentials, purpose and commitment to mitigating dual-use risks. Source data are provided with this paper.

Code availability

Code is available at <https://osf.io/vx5su/>.

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Author contributions H.L., G.C., B.L., G.P. and D.G.R. conceived of the research; H.L., G.P. and D.G.R. designed the US and Canadian experiments; B.L., J.P.W. and D.G.R. designed the Massachusetts ballot experiment; H.L., G.C., G.P. and D.G.R. designed the Polish experiment; H.L., G.C., B.L. and D.G.R. analysed the data. H.L., G.P. and D.G.R. wrote the paper, with input from G.C., B.L., J.P.W., A.J.B. and T.C. All authors approved the final manuscript.

Competing interests Other research by D.G.R. has been funded by Google and Meta. The other authors declare no competing interests.

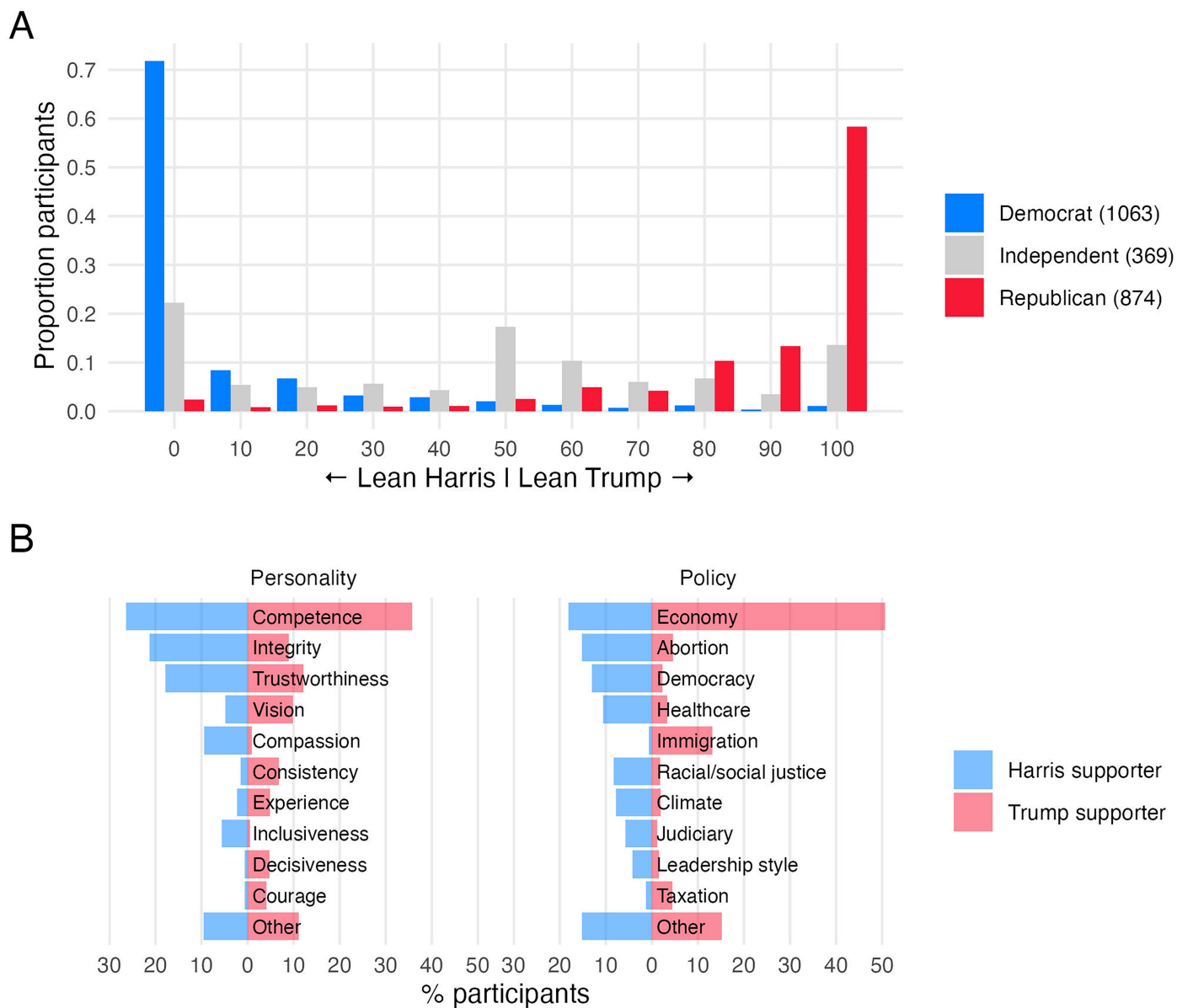
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-025-09771-9>.

Correspondence and requests for materials should be addressed to Gordon Pennycook or David G. Rand.

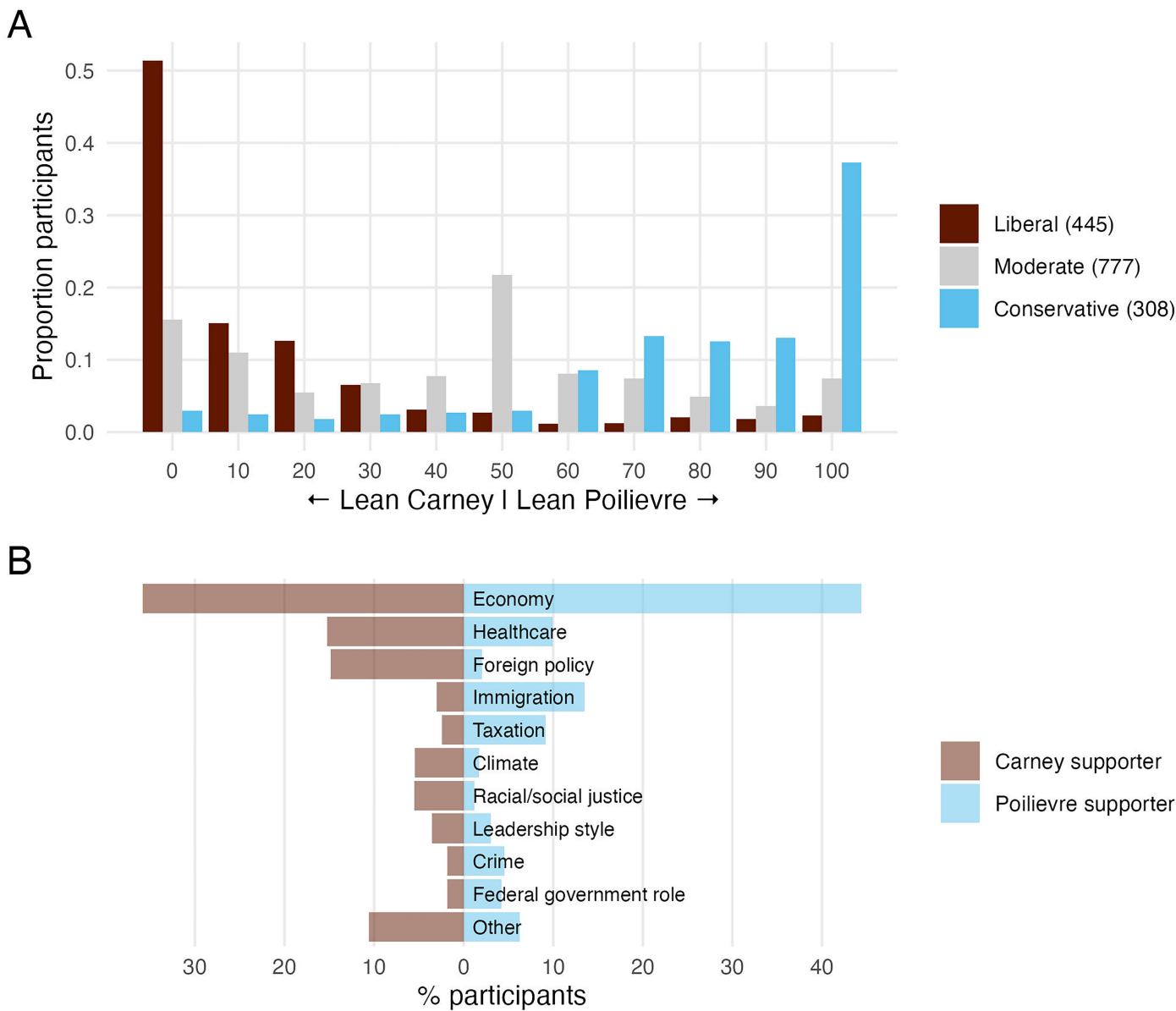
Peer review information Nature thanks Ho-Chun Herbert Chang, Alessandro Nai, Chiara Vargiu and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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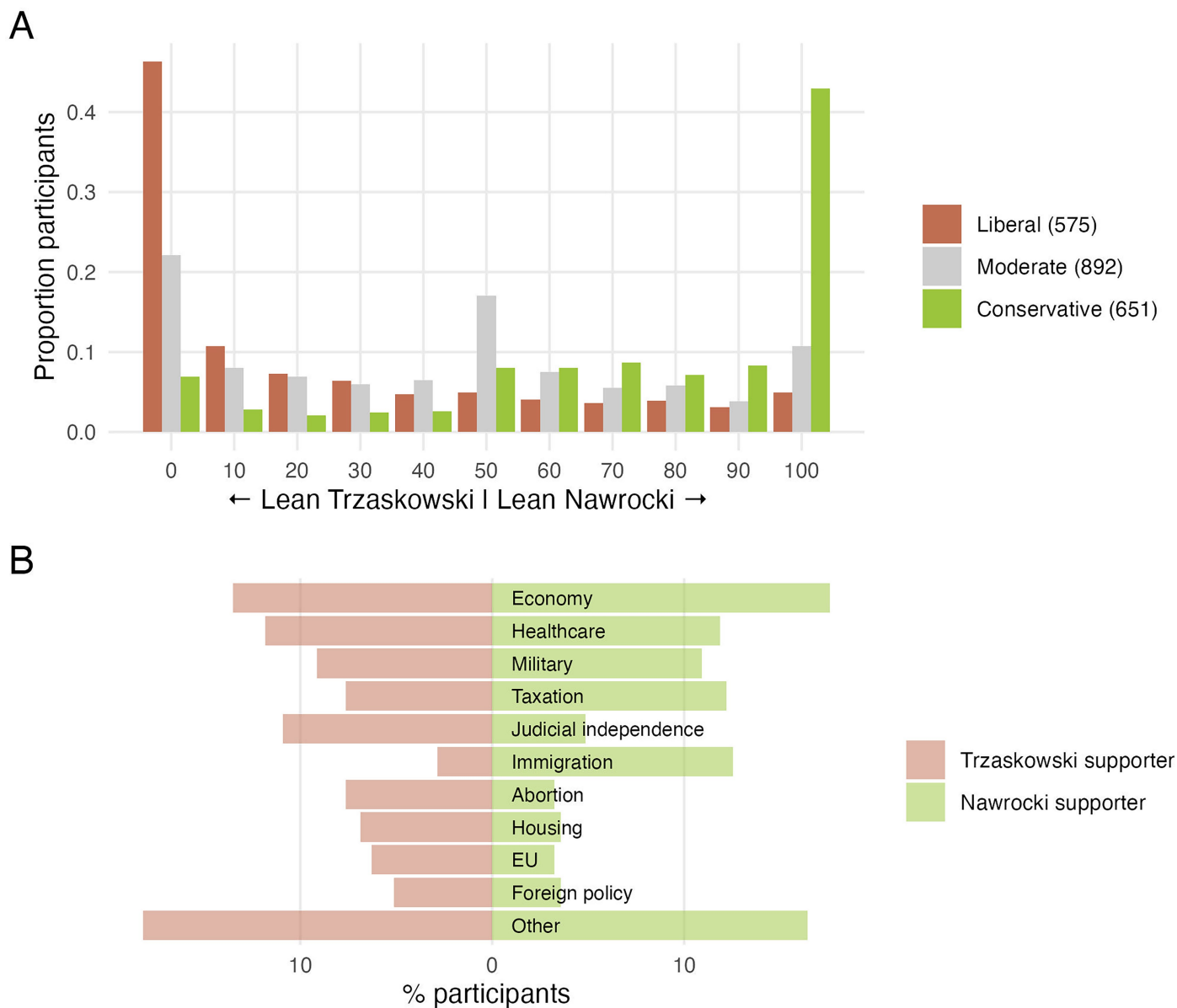
Extended Data Fig. 1 | Top-importance issues varied markedly across parties among the highly polarized participants in the U.S. presidential election experiment. (A) Distribution of pre-treatment candidate preference by party in our sample. Partisans entered the study with highly polarized views of the candidates. Thus, there was limited room for the pro-Harris AI to persuade Harris supporters or the pro-Trump AI to persuade Trump supporters. (B) Participants were randomized to indicate either the candidate's personal characteristic or

policy issue that was the most important to them (selected from pre-determined lists of 19 personal characteristics and 19 policy issues). Here we show the fraction of Harris and Trump supporters selecting each option (the top 10 are shown, while the remaining 9 are collapsed into the "Other" category). We see marked differences between Harris and Trump supporters in terms of their top-priority personality characteristics and policy issues. This heterogeneity emphasizes the potential value of tailored persuasion via AI dialogues.



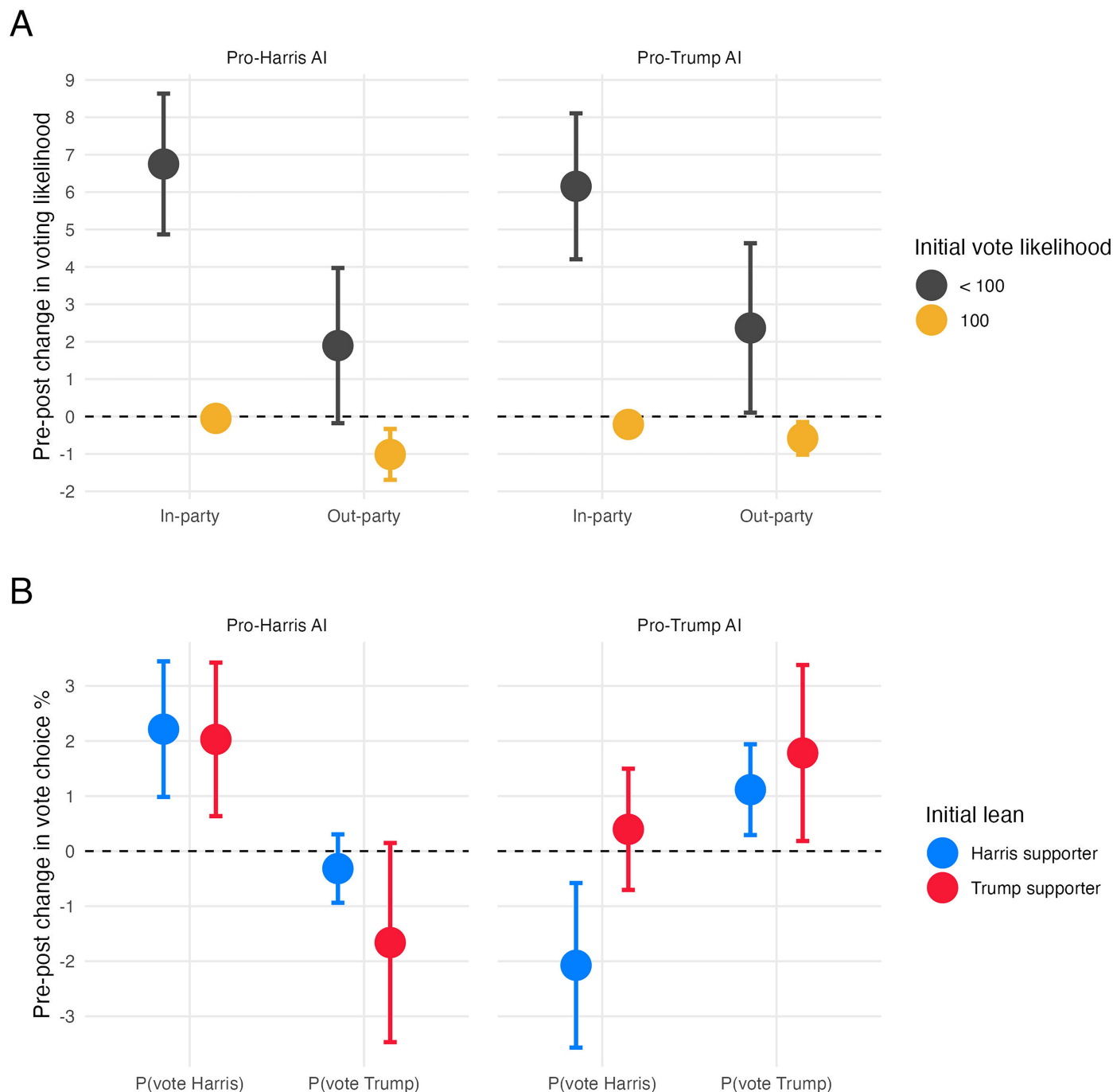
Extended Data Fig. 2 | Top-importance issues varied markedly across parties among the highly polarized participants in the Canadian federal election experiment. (A) Distribution of pre-treatment candidate preference by party in our sample. Partisans entered the study with polarized views of the candidates. (B) Participants indicate the policy issue that was the most important to them

(selected from a pre-determined list of 17 policy issues). Here we show the fraction of Carney and Poilievre supporters selecting each option (the top 10 are shown, while the remaining are collapsed into the “Other” category). We see differences between Carney and Poilievre supporters in terms of their top-priority policy issues.



Extended Data Fig. 3 | Top-importance issues among participants in the Polish presidential election experiment. (A) Distribution of pre-treatment candidate preference by party in our sample. Partisans entered the study with polarized views of the candidates. (B) Participants indicate the policy issue that

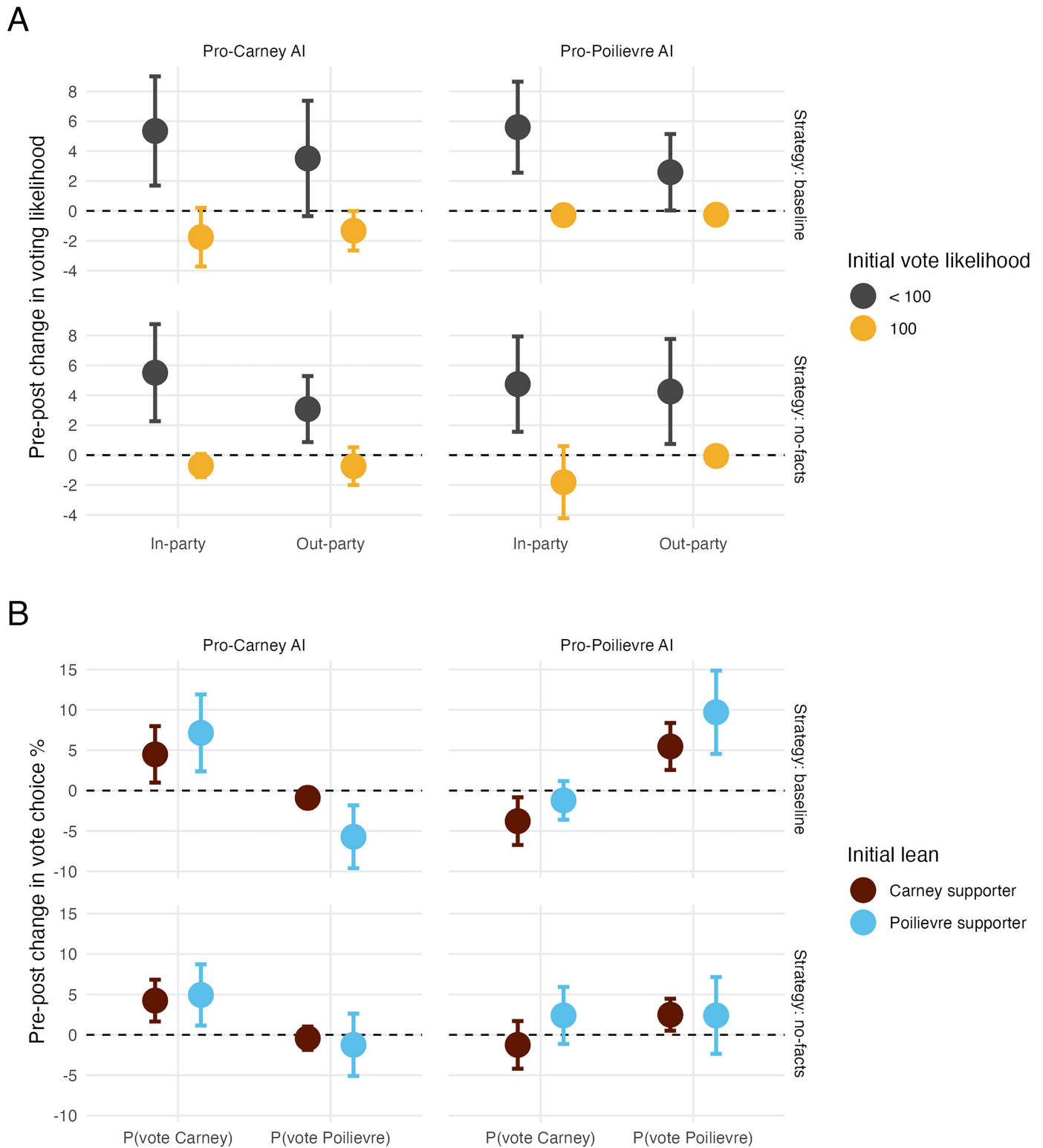
was the most important to them (selected from a pre-determined list of 18 policy issues). Here we show the fraction of Trzaskowski and Nawrocki supporters selecting each option (the top 10 are shown, while the remaining are collapsed into the “Other” category).



Extended Data Fig. 4 | The AI conversations changed participants' voting likelihood and vote choice in the U.S. presidential election experiment.

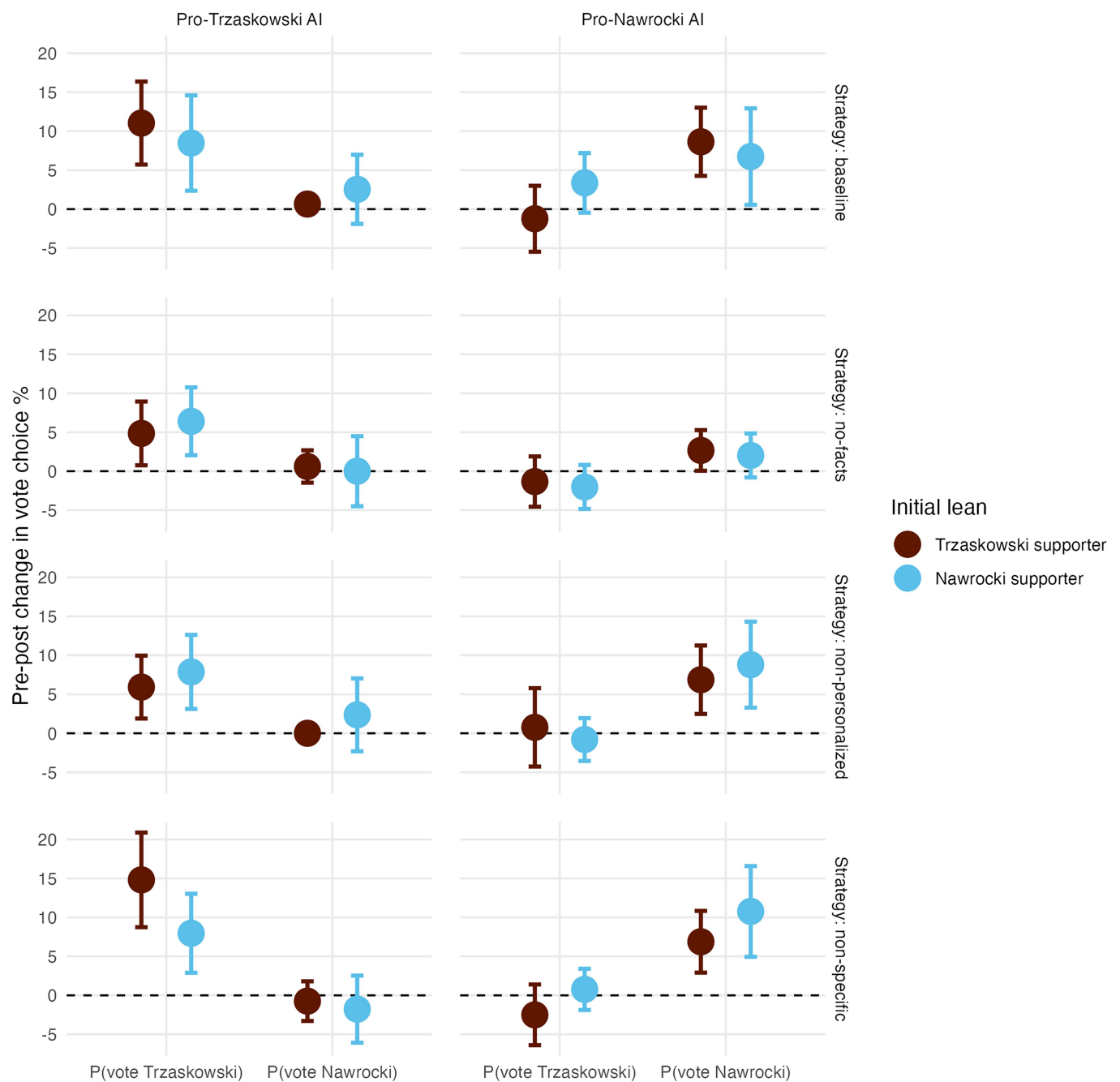
(A) Shown is the pre-to-post change in voting likelihood (0–100 scale) by randomly assigned conditions and participants' pre-treatment voting likelihood (median split into <100 versus 100) (preregistered analysis). Both pro-Harris and pro-Trump treatments increased voting intentions among in-party participants whose pre-treatment voting intention was not already at ceiling (100%). The pre-to-post decrease in voting likelihood among out-party participants initially

at ceiling likely reflect regression to the mean (as they can only down from 100%). (B) Shown is the pre-to-post change in vote choice (categorical outcome based on who participants report they would vote for if the election happened today) (*post hoc* analysis). The left half of each panel shows the fraction of people choosing Harris, whereas the right half of each panel shows the fraction of people choosing Trump. The pro-Harris AI and pro-Trump AI increased the probability of voting for Harris versus Trump, respectively. Error bars indicate 95% confidence intervals of the mean. See SI Section 2.1 for details.



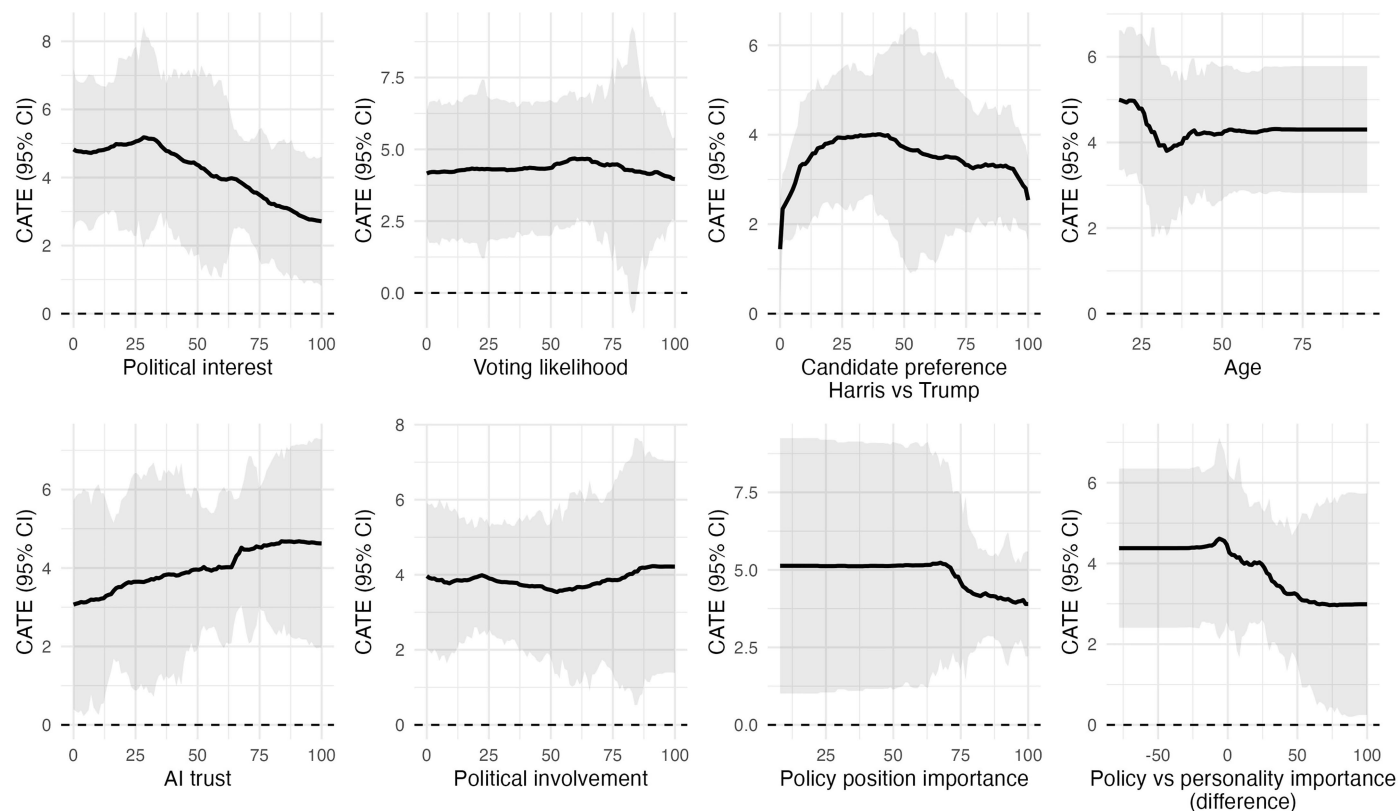
Extended Data Fig. 5 | The AI conversations changed participants' voting likelihood and vote choice in the Canadian federal election experiment (preregistered analysis). (A) Shown is the pre-to-post change in voting likelihood (0–100 scale) by randomly assigned conditions and participants' pre-treatment voting likelihood (median split into <100 versus 100). Both pro-Carney and pro-Poillievre treatments increased voting intentions among in-party participants whose pre-treatment voting intention was not already at ceiling (100%). The pre-to-post decrease in voting likelihood among participants

initially at ceiling likely reflect regression to the mean (as they can only down from 100%). (B) Shown is the pre-to-post change in vote choice (categorical outcome based on who participants report they would vote for if the election happened today). The left half of each panel shows the fraction of people choosing Carney, whereas the right half of each panel shows the fraction of people choosing Poillievre. The pro-Carney AI and pro-Poillievre AI increased the probability of voting for Carney versus Poillievre, respectively. Error bars indicate 95% confidence intervals of the mean. See SI Section 2.2 for details.



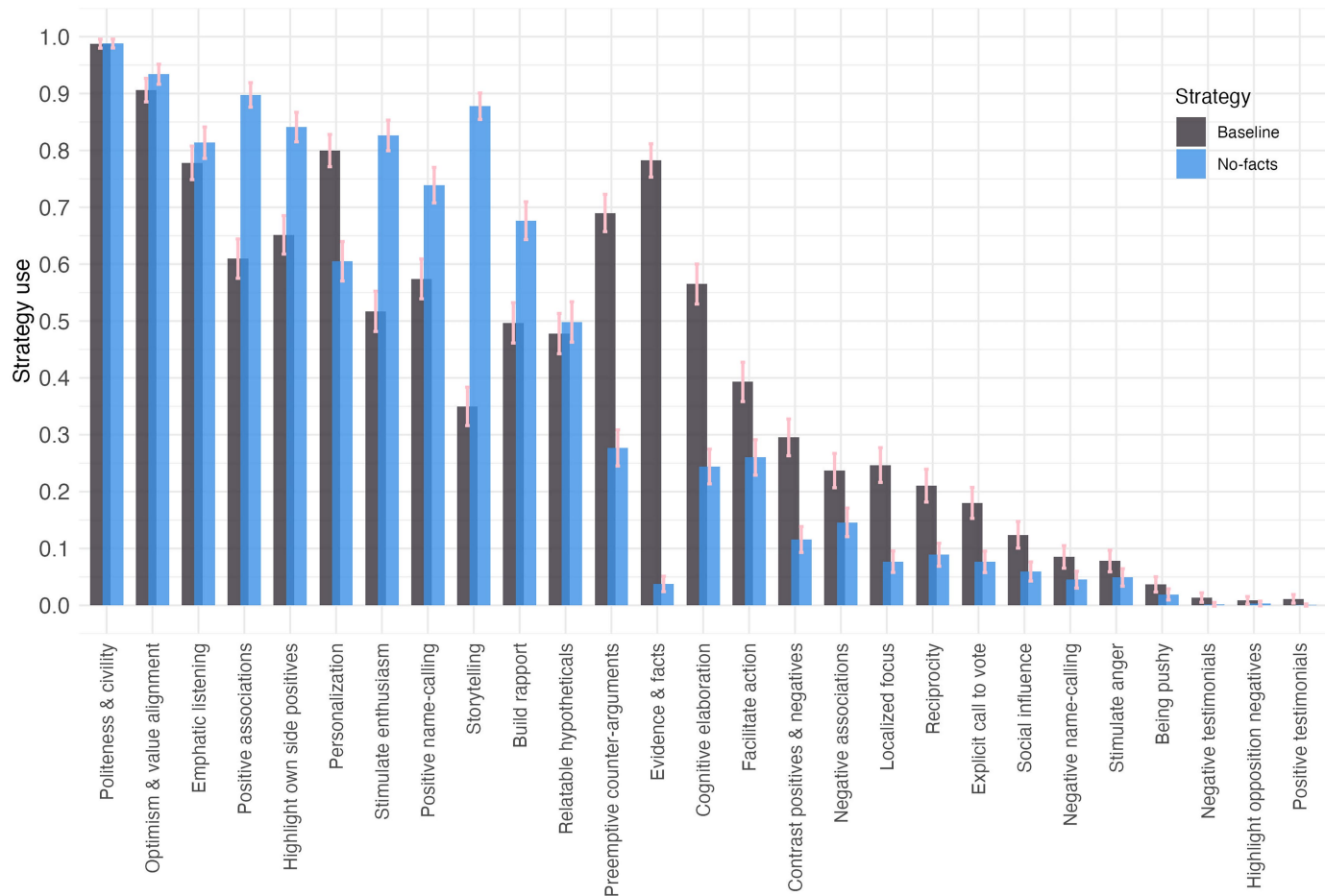
Extended Data Fig. 6 | The AI conversations changed participants' vote choice in the Polish presidential election experiment (preregistered analysis). Shown is the pre-to-post change in vote choice (categorical outcome based on who participants report they would vote for if the election happened today). The left half of each panel shows the fraction of people choosing Trzaskowski,

whereas the right half of each panel shows the fraction of people choosing Nawrocki. The pro-Trzaskowski AI and pro-Nawrocki AI increased the probability of voting for Trzaskowski versus Nawrocki, respectively. Error bars indicate 95% confidence intervals of the mean. See SI Section 2.3 for details.



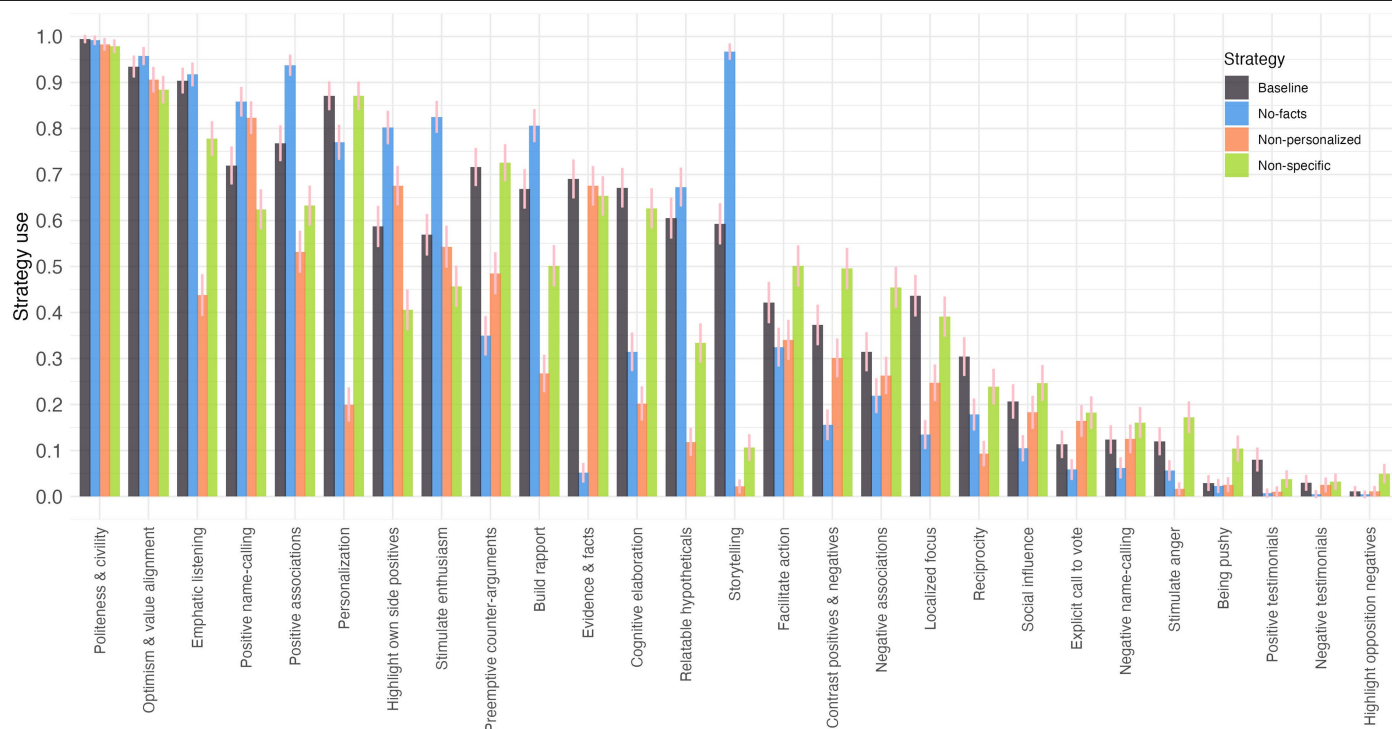
Extended Data Fig. 7 | Causal forest conditional average treatment effects (CATE) on candidate preference in the U.S. presidential election experiment (*posthoc* analysis). Shown are the partial dependence plots for pre-treatment covariates that are considered important (i.e., weighted sum of how many times a feature was split on at each depth in the causal forest) for predicting post-treatment candidate preference in the training set, ordered left to right by importance. A covariate is considered “important” if its importance value is higher than the mean importance across all covariates. The CATEs are estimated on a holdout testing dataset; for each covariate, we hold all other covariates at their median values. For example, pre-treatment candidate preference has an inverted-U relationship with post-treatment candidate preference, such that

more moderate participants (with no strong preference for Harris [0] or Trump [100]) show stronger treatment effects. Participants with higher political interest and involvement have weaker (but still positive) treatment effects. Error bars indicate 95% confidence intervals of the mean. Pre-treatment covariates used for model fitting: candidate preference, voting likelihood, conversation focus (personality or policy), age, income, education, belief in God, partisanship, political orientation, social conservatism, economic conservatism, area lived in, gender, ethnicity, general trust in AI, political interest, political involvement, warmth toward Democrats (feeling thermometer), warmth toward Republicans, importance of considering policy issue, importance of considering personality, policy versus personality consideration importance (difference).



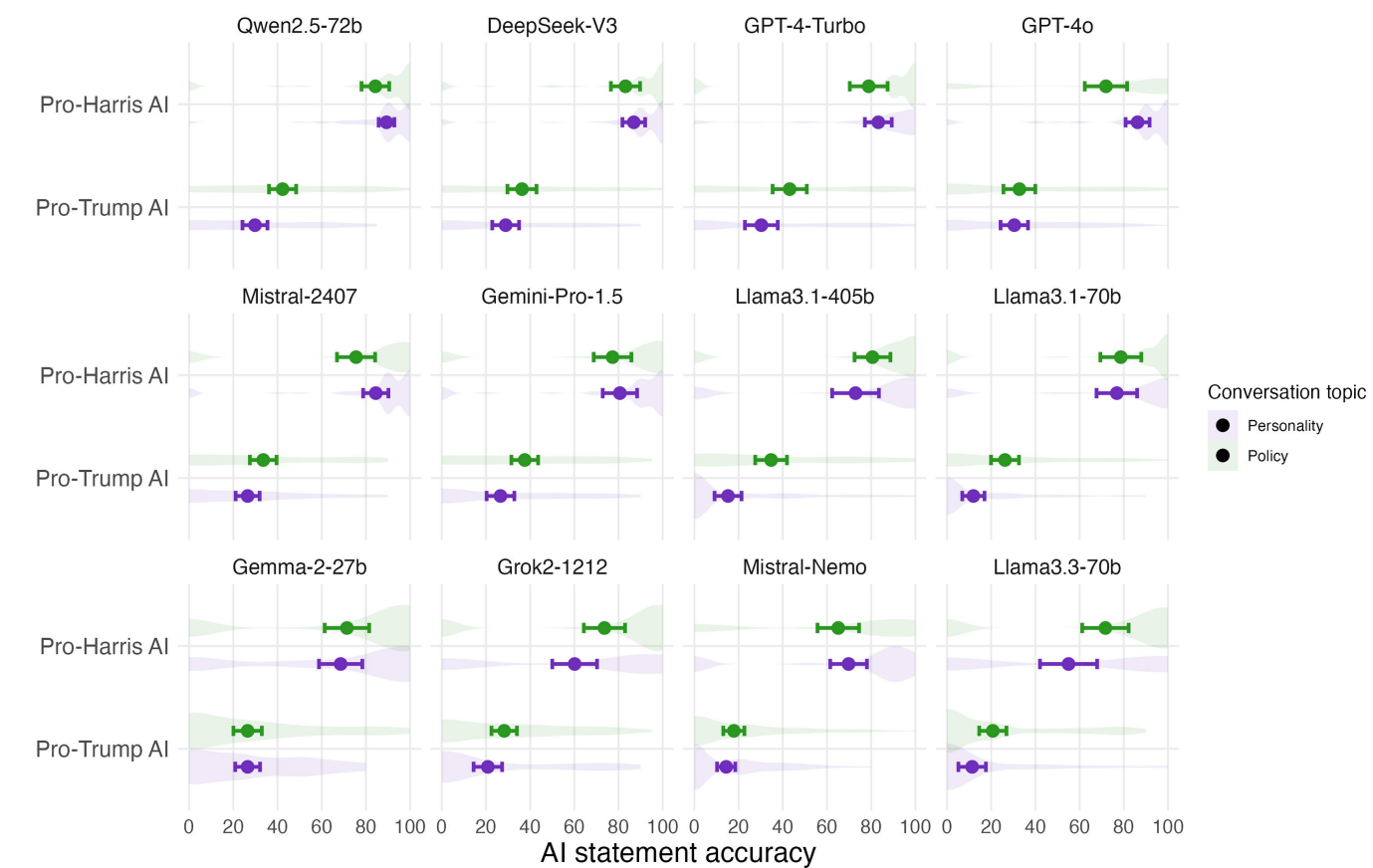
Extended Data Fig. 8 | The AI models used different persuasion strategies in the Canadian federal election experiment. Shown is the average use of 27 strategies across conversations. For each conversation, two different AI models rated the use of each strategy using a 3-level scale (0, 0.5, 1) and those two ratings

were then averaged. The AI model in the no-facts condition used much less evidence and facts than the AI model in the baseline condition. Error bars indicate 95% confidence intervals of the mean.



Extended Data Fig. 9 | The AI models used different persuasion strategies in the Polish presidential election experiment. Shown is the average use of 27 strategies across conversations. For each conversation, two different AI models rated the use of each strategy using a 3-level scale (0, 0.5, 1) and those two ratings were then averaged. The AI model in the no-facts condition used

much less evidence and facts than the AI model in the baseline condition. The AI model in the non-personalized condition did not personalize its responses to the participants as much. Error bars indicate 95% confidence intervals of the mean.



Extended Data Fig. 10 | AI model statement accuracy varied across models and experimental conditions (exploratory post hoc robustness check). Shown are the distributions with means and 95% CIs of accuracy ratings for simulated statements generated by 12 open-source and proprietary AI models for the candidate preference experiment. For all 12 models, the pro-Trump AI

made more inaccurate statements than the pro-Harris AI, particularly when discussing personal characteristics. The accuracy ratings were determined by Perplexity, a large language model that can access real-time information from the internet. Error bars indicate 95% confidence intervals of the mean.

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☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
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☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
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☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

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Software and code

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Data collection	Qualtrics survey software (August 2024) was used to design and administer the surveys and collect data. Vegapunk (v1.0.0) was used to implement the conversation interface in Qualtrics.
Data analysis	Python (3.13.1) and R (version 4.4.1). LLMs: OpenAI GPT (4o), Deepseek (V3), Perplexity (Sonar) for AI-based fact-checking.

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Data Availability
Due to the sensitive nature of this research—namely, the dual-use potential associated with the infrastructure and data used for political persuasion—we do not publicly share raw conversation transcripts or the bespoke LLM code. Release of these materials could facilitate misuse, such as the development or fine-tuning of

models for unethical persuasion. These materials may be made available from the authors upon reasonable request through Harvard Dataverse at <https://doi.org/10.7910/DVN/DODEXZ>. Requests will be evaluated on an individual basis, considering the researcher's credentials, purpose, and commitment to mitigating dual-use risks.

Code Availability

Code is available at <https://osf.io/vx5su/>.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

Sex and gender were not considered in the study design. Gender data has been collected and determined based on self-reporting. Consent has been obtained for sharing of individual-level data. We did not perform sex- and gender-based analyses because they were not relevant to the research questions. Gender numbers:

- US election study: 1,215 females, 1,038 males, and 53 who selected another gender option
- Ballot measure study: 266 females, 213 males, and 22 who selected another gender option
- Canada election study: 718 females, 797 males, and 15 who selected another gender option
- Polish election study: 904 females, 1,196 males, and 18 who selected another gender option

Reporting on race, ethnicity, or other socially relevant groupings

We collected standard demographic variables such as political preference/orientation (7-point scales: strongly Democratic to Strongly Republican; very liberal to very conservative), social/economic conservatism, age, income, ethnicity, and education through self-report. We did not use these variables as proxies for other social constructs. Our studies involve randomization/experimentation, so confounding variables are unlikely to significantly affect our conclusions.

Population characteristics

See above.

Recruitment

Participants were recruited through CloudResearch Connect, Prolific, Lucid and PureSpectrum. Because our studies were related to elections and political preferences, we aimed to recruit roughly equal numbers of Democrat/liberal and Republican/conservative participants to ensure our results are generalizable and valid. Participants may have self-selected into taking part in online surveys/research and may differ systematically from the general population—for example, being more politically engaged, or having stronger opinions—this self-selection could introduce bias that limits the external validity of the findings. We conducted post hoc causal forest analysis to examine whether different subgroups might show different responses to our experimental manipulations, which helps assess the robustness of our conclusions across demographic and ideological lines (e.g., the treatment effects are actually higher among participants who are less politically engaged/interested).

Ethics oversight

MIT Committee on the Use of Humans as Experimental Subjects (protocol E-5990, E-6748, E-6757)

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description

Study designs (quantitative and experimental) are detailed in the Methods section. The studies involve asking participants to engage in a multi-turn conversation with an AI/LLM chatbot about different topics/issues, and seeing if participants' preferences change after interacting with the chatbot.

Research sample

The studies recruited American/Canadian/Polish participants from CloudResearch Connect, Prolific, Lucid, or PureSpectrum. Because our studies were related to elections and political preferences, we aimed to recruit roughly equal numbers of Democrat/liberal and Republican/conservative participants to ensure our results are generalizable and valid. The samples may not be fully representative (despite our attempts to ensure representativeness) because such online participant pools tend to slightly overrepresent liberal individuals.

US experiment: We preregistered a target sample of 2,400 American participants, recruiting from CloudResearch Connect and Prolific. In total, 2,457 participants started the survey, and 2,306 (1,063 Democrats, 874 Republicans, 369 Independents) completed the survey (see Tables 1 and 2). This sample (mean age = 38.13 [SD = 12.18]) included 1,215 females, 1,038 males, and 53 who selected another gender option. Sample characteristics (SDs in square brackets): AI trust = 53.83 [28.02] (0-100 scale), economic conservatism = 3.00 [1.29] (1-5 scale), social conservatism = 2.63 [1.30] (1-5 scale), belief in God = 4.09 [2.76] (0-8 scale), political interest = 60.91 [28.93] (0-100 scale) (see Fig. S1 for distributions of pre-treatment covariates).

Ballot measure experiment: We preregistered a target sample of 500 Massachusetts residents. We initially pre-registered that we

would aim to recruit 250 Democrats and 250 Republicans, but given the tight timeline for subject recruitment and the limited pool of Massachusetts residents available, we decided not to implement partisanship quotas. We recruited participants from both Prolific and CloudResearch Connect. In total, 522 participants began the survey. As per our pre-registration, participants who failed to complete the pre-treatment free-text explanation of their position on the ballot measure were dropped. All who started the survey completed the pre-treatment free-text explanation, so 522 participants were assigned to treatment. Of these, another 21 dropped out during the treatment; the dropout rate did not differ significantly across conditions, $\chi^2(2) = 0.49$, $p = .784$ (see SI Section S10 Fig. S30). This leaves us with a final set of 501 participants (291 Democrats, 85 Republicans, 125 Independents; mean age = 38.73 [SD = 13.74]; 266 females, 213 males, and 22 who selected another gender option). Sample characteristics (SDs in square brackets): AI trust = 4.12 [1.57] (1-7 scale), economic conservatism = 3.42 [1.75] (1-7 scale), social conservatism = 2.76 [1.62] (1-7 scale) (see Fig. S2 for distributions of pre-treatment covariates).

Canadian experiment: We preregistered a target sample of 4,000 Canadians, recruiting from four recruitment platforms (CloudResearch Connect, Lucid, Prolific, PureSpectrum) to ensure we could recruit as many participants as possible in the five days leading up to the Canadian elections on April 28, 2025. In total, 1,694 participants began the survey, and 1,530 completed the survey (850 Liberals, 490 Conservatives, 211 neither Liberal nor Conservative). This sample (mean age = 41.41 [SD = 16.67]) included 718 females, 797 males, and 15 who selected another gender option. Sample characteristics (SDs in square brackets): AI trust = 42.41 [16.67] (0-100 scale), economic conservatism = 2.89 [1.17] (1-5 scale), social conservatism = 2.51 [1.20] (1-5 scale), belief in God = 4.60 [2.71] (0-8 scale), political interest = 58.70 [28.16] (0-100 scale) (see Fig. S3 for distributions of pre-treatment covariates).

Polish experiment: We preregistered a target sample of 4,000 Poles recruited from Lucid. In total, 2,380 participants began the survey, and 2,118 completed the survey (mean age = 39.41 [SD = 20.24]; 904 females, 1,196 males, and 18 selected another gender option). Sample characteristics (SDs in square brackets): AI trust = 58.27 [27.45] (0-100 scale), economic conservatism = 2.93 [1.06] (1-5 scale), social conservatism = 2.76 [0.98] (1-5 scale), belief in God = 4.98 [2.53] (0-8 scale), political interest = 64.24 [26.10] (0-100 scale) (see Fig. S4 for distributions of pre-treatment covariates).

Sampling strategy	Sample sizes were based on previous studies of this nature (see also response to Research sample above for details), the maximum amount of money we wished to spend on each study, and the amount of time we had leading up to each election (given the research questions, we had to end data collection before election day). We had no strong basis for an a prior estimated effect size and such power analyses are often uninformative (http://datacolada.org/4).
Data collection	We used Qualtrics survey software to collect data from online recruitment platforms. The researcher was not present with the participant when data collection happened (so the researcher was blinded to the experimental condition). Given data collection happened online/remotely, we cannot determine whether someone else was with the participant.
Timing	The US experiment was conducted from Aug 22 to Nov 2, 2024. The ballot measure experiment was conducted from Oct 30 to Nov 5, 2024. The Canadian experiment was conducted from April 23 to April 28, 2025. The Polish experiment was conducted from May 7 to May 16, 2025.
Data exclusions	We excluded participants for inattentiveness before the study using an open-ended text response. In addition, in the ballot measure study, participants who failed to complete the pre-treatment free-text explanation of their position on the ballot measure were excluded. US experiment: 2,457 participants started the survey, and 2,306 (1,063 Democrats, 874 Republicans, 369 Independents) completed the survey (see Tables 1 and 2). Ballot measure experiment: In total, 522 participants began the survey. As per our pre-registration, participants who failed to complete the pre-treatment free-text explanation of their position on the ballot measure were dropped. All who started the survey completed the pre-treatment free-text explanation, so 522 participants were assigned to treatment. Of these, another 21 dropped out during the treatment; the dropout rate did not differ significantly across conditions. Canadian experiment: 1,694 participants began the survey, and 1,530 completed the survey (850 Liberals, 490 Conservatives, 211 neither Liberal nor Conservative). Polish experiment: 2,380 participants began the survey, and 2,118 completed the survey.
Non-participation	151 of 2,457 participants dropped out in the US experiment; 21 of 522 participants dropped out in the ballot measure study; 164 of 1,694 participants dropped out in the Canadian experiment; 262 of 2,380 participants dropped out in the Polish experiment. We reported analyses in the SI showing that there was no evidence for differential attrition/dropout across experimental conditions. Because of the online nature of the studies, we are unable to collect data on why the participants dropped out.
Randomization	Randomization was implemented using Qualtrics survey software.

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Materials & experimental systems

Methods

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☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☐	☒ Dual use research of concern
☒	☐ Plants

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Dual use research of concern

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Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

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☐	☒ National security
☒	☐ Crops and/or livestock
☒	☐ Ecosystems
☐	☒ Any other significant area

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Hazards

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☒	☐ Enable evasion of diagnostic/detection modalities
☒	☐ Enable the weaponization of a biological agent or toxin
☐	☒ Any other potentially harmful combination of experiments and agents

Other combinations

Precautions and benefits

Biosecurity precautions

Biosecurity oversight

Benefits

Communication benefits

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Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.